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Chapter 9. Prescriptive Analytics: Causal Machine Learning

Data Science for Business

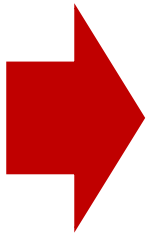
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Prof. dr. Wouter Verbeke

Prof. dr. Bart Baesens



Course schedule



BLOCK2			
6	26/10/2026	Chapter 6. Predictive Analytics - Artificial Neural Networks	
6	27/10/2026	**No class~~, video-lecture: Chapter 7. Predictive Analytics - Forecasting~~**	
7	02/11/2026	**No class, university holiday**	
7	03/11/2026	Chapter 9. Prescriptive Analytics - Causal Machine Learning	
8	09/11/2026	**No class~~, video-lecture: Chapter 8. Descriptive Analytics~~**	
8	10/11/2026	Chapter 9. Prescriptive Analytics - Causal Machine Learning	**Report 1 deadline**
9	16/11/2026	Chapter 10. Prescriptive Analytics - Profit-driven Analytics	
9	17/11/2026	Guest Lecture #1: Toon Vanderschueren, Ph.D., Causal ML @Telenet	**Presence expected**
	17/11/2026	Assignment 2	**Report 2 > 14/12**
10	23/11/2026	Chapter 11. Prescriptive Analytics - Multi-armed Bandits & Reinforcement Learning	
10	24/11/2026	Guest Lecture #2: Dr. Véronique Van Vlasselaer, Ph.D., Business Analytics @SAS	**Presence expected**
BLOCK3			
11	30/11/2026	Chapter 12. Natural Language Processing	
11	01/12/2026	Chapter 12. Natural Language Processing	
12	07/12/2026	Chapter 13. Large Language Models & AI Agents	
12	08/12/2026	Chapter 13. Large Language Models & AI Agents	
13	14/12/2026	Feedback Assignment 2 Sample Exam Questions	**Report 2 deadline**
13	15/12/2026	Guest Lecture #3: To be confirmed	**Presence expected**

Prescriptive Analytics?

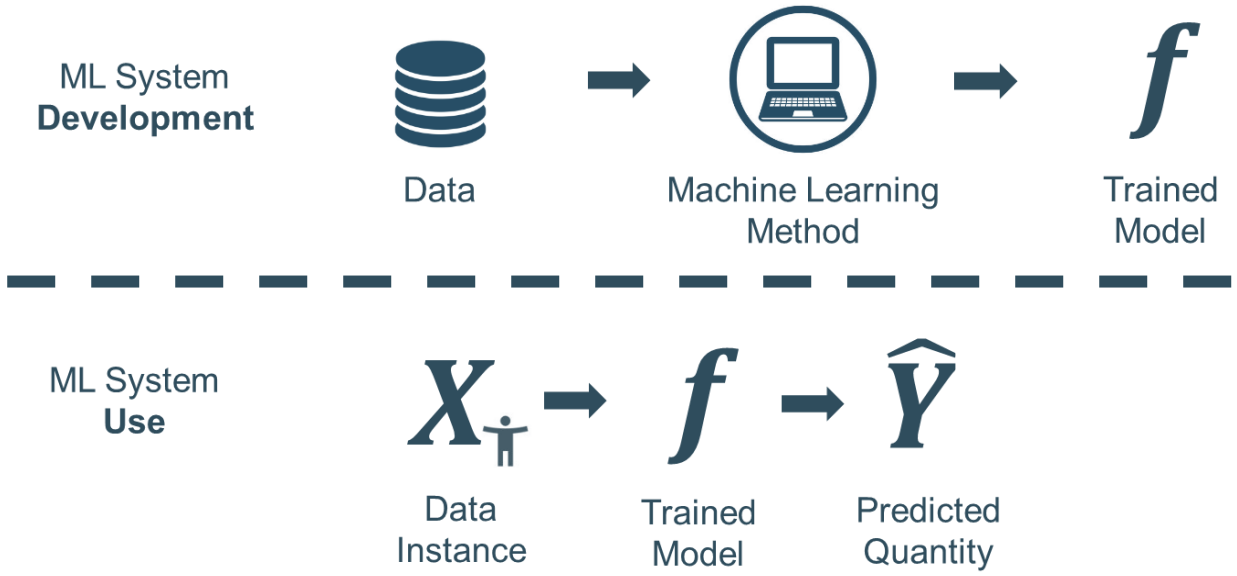
- Cf. Chapter 1

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Machine learning for business decision-making

A prediction is not a decision!

2 ***How can machine learning be used for business decision-making?***

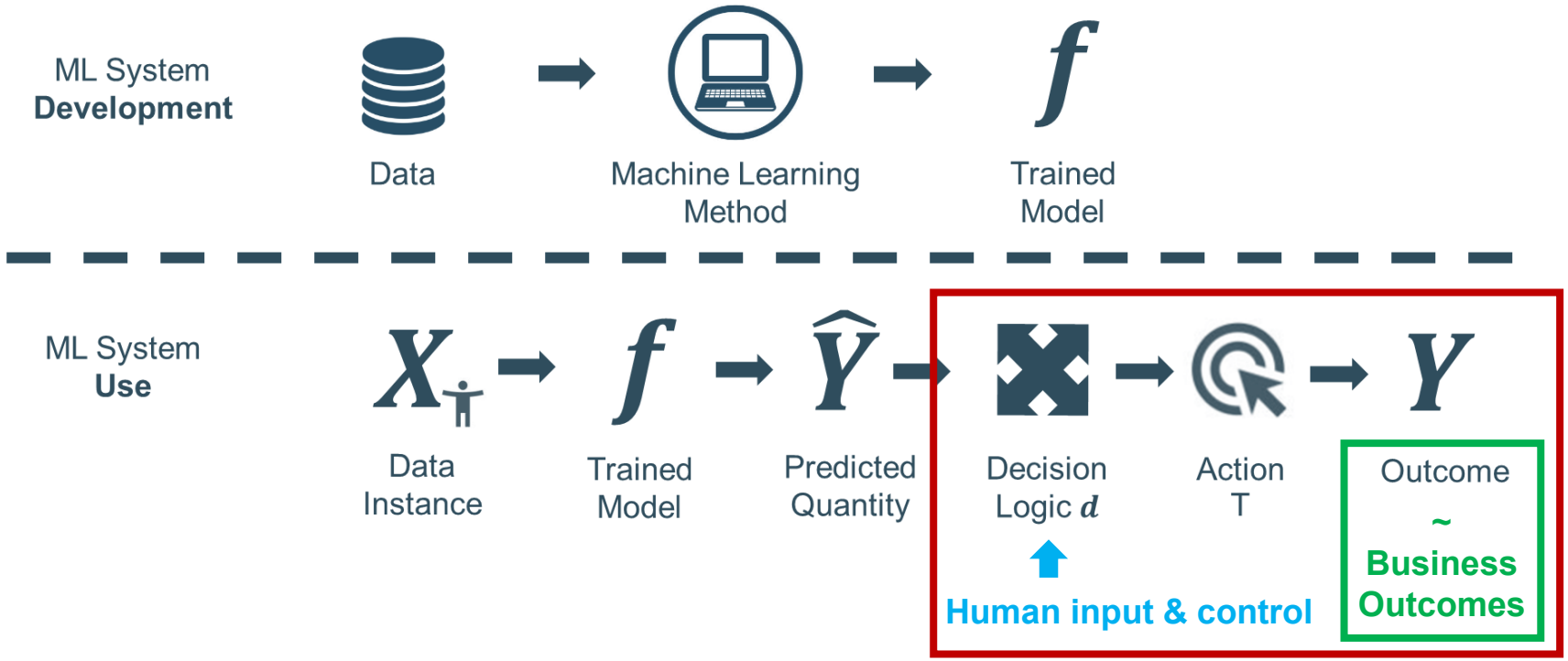


Credits for figure to prof. Foster Provost

Machine learning for business decision-making

A prediction is not a decision! → **Decision logic d with $T = d(\hat{Y}, \ldots)$** (~policy, optimization, rules, …)

- Determines **how predictions result in actions or decisions**
- Taking into account **business objectives, constraints and requirements**
- **Benefits in integrating with ML System development? How to?**



Business objectives
- Profit / Loss
- Risk / Uncertainty
- Stability

Business constraints
- Budget / Capacity
- Operational
- Regulatory

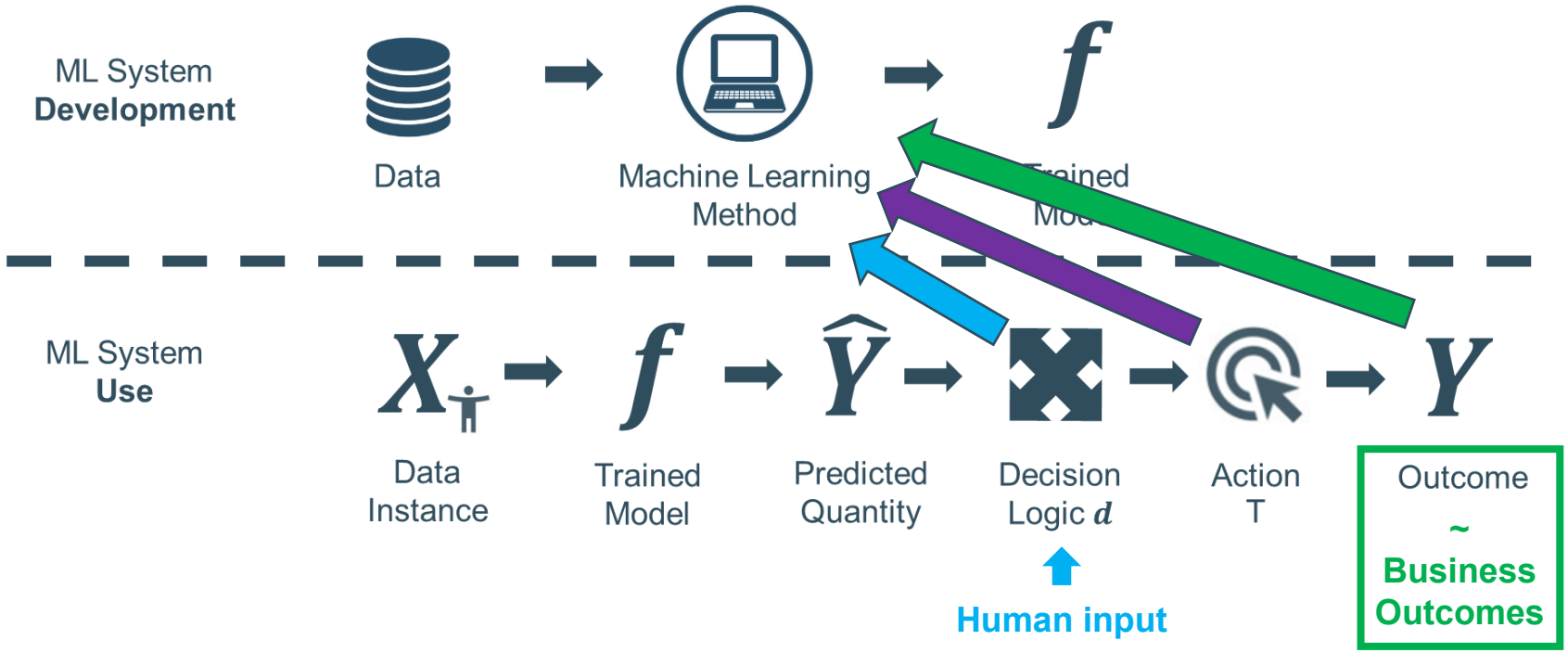
Business requirements
- Explainability
- Stability (cf. infra)

Credits for figure to prof. Foster Provost

Machine learning for business decision-making

A prediction is not a decision! → **Decision logic d with $T = d(\hat{Y}, \dots)$** (~policy, optimization, rules, …)

- Determines **how predictions result in actions or decisions**
- Taking into account **business objectives, constraints and requirements**
- **Benefits in integrating with ML System development? How to?**



Decision-centric learning
Predict-then-optimize
Predict-and-optimize
…

Causal machine learning

Reinforcement learning
Prescriptive analytics
…

Cost-sensitive learning
Profit-driven analytics
…

Credits for figure to prof. Foster Provost

Learning objectives

- At the end of this chapter, students should be able to:
  - Understanding the **difference** between **descriptive, predictive and prescriptive data analytics**, and understanding causal machine learning as an approach to the latter
  - Understanding **challenges** in applying causal machine learning
  - Being able to identify **opportunities** for applying causal machine learning
  - Understanding **basic approaches** for developing causal machine learning
  - Understanding **basic performance measures** for evaluating and monitoring causal machine learning

Outline

1. Case: customer churn
2. Causal inference
3. Challenges
4. Methods
5. Evaluation
6. Cases
7. Concluding remarks

1. Case: customer churn

Exercise

Does this look like a good solution to you?

What business problem does it solve?

Case: V-Bank

- V-Bank segments its customers in 4 segments: bronze, silver, gold and platinum, depending on the amount invested and the uptake of financial products and services.

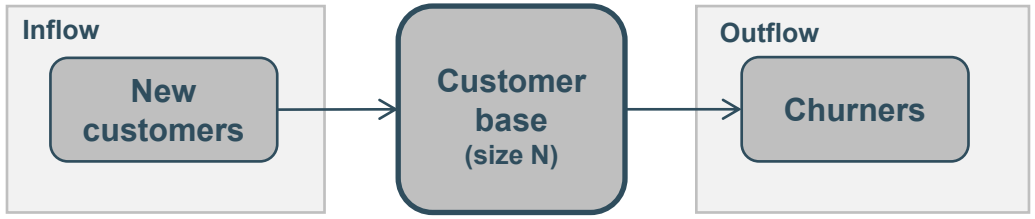
- A data scientist in your team proposes to develop a predictive model to identify customers that will upgrade to a higher segment over the next 24 months, in order to target them with a campaign to up their investment level.

- To this end, she proposes to use data from the past five years, during which about 10% of the customers was upgraded from one segment to the next.

Case: customer churn

Analytics to the rescue?

- Customer churn: outflow of customers from the customer base



- Losing customers leads to a **direct revenue loss**
- **Attracting new customers:** 5 to 6 times more expensive than keeping existing clients satisfied through **retention campaigns**
  - Advertisement
  - Appealing conditions/offer (U.S.: 500$ to generate 50$/month client)

Colgate, M. R., & Danaher, P. J. (2000). Implementing a customer relationship strategy: The asymmetric impact of poor versus excellent execution. *Journal of the Academy of Marketing Science*, *28*(3), 375-387.

Case: customer churn

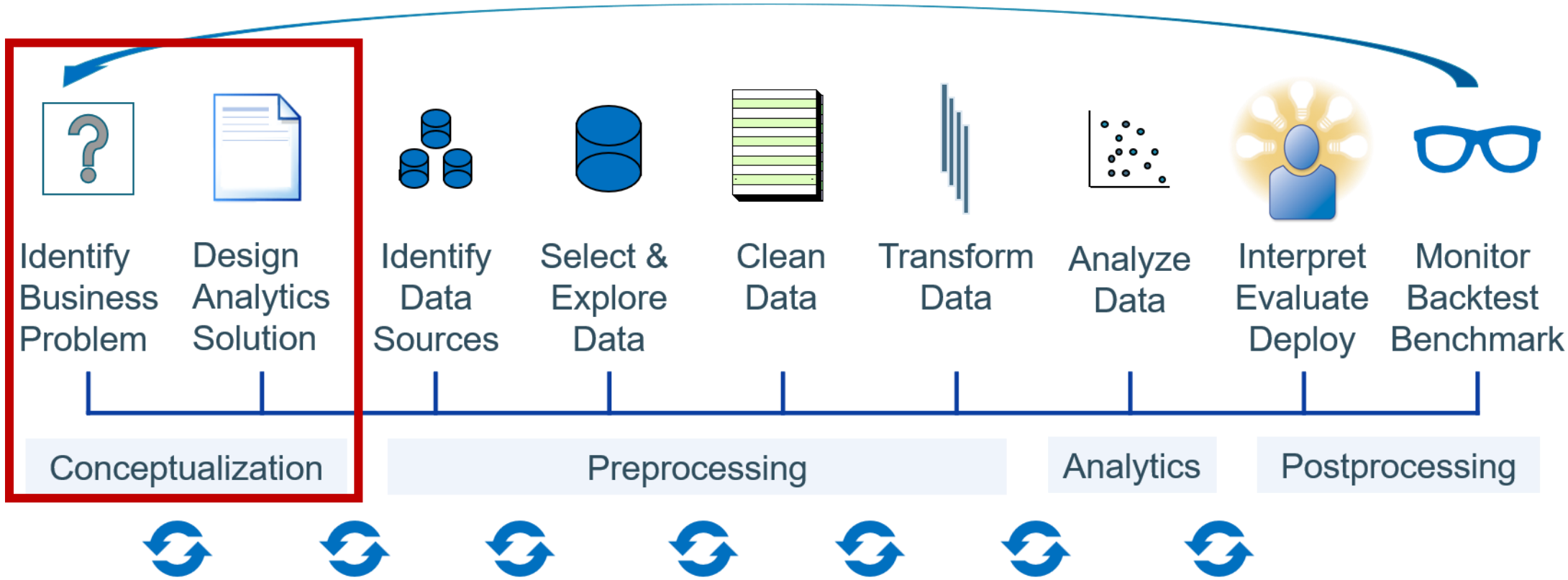
Analytics to the rescue?

- Ascarza et al. (2018): comprehensive framework of questions for designing effective retention strategies
  - "Who is at risk and why?"
  - "Whom should we target and with what incentives?"
  - "When do we target?"
  - "What can we gain?"
- Data-driven solution: **customer churn prediction models**
  - Identification of churners and their motivations allows to:
    - **Target** future churners in retention campaigns
    - Increase **efficiency** of such campaigns
  - Small improvement in customer retention may yield significant profit increase

Ascarza, E. et al. (2018). In pursuit of enhanced customer retention management: Review, key issues, and future directions. *Customer Needs and Solutions*, *5*(1), 65-81.

Case: customer churn

- Practical steps in elaborating a customer churn prediction model:



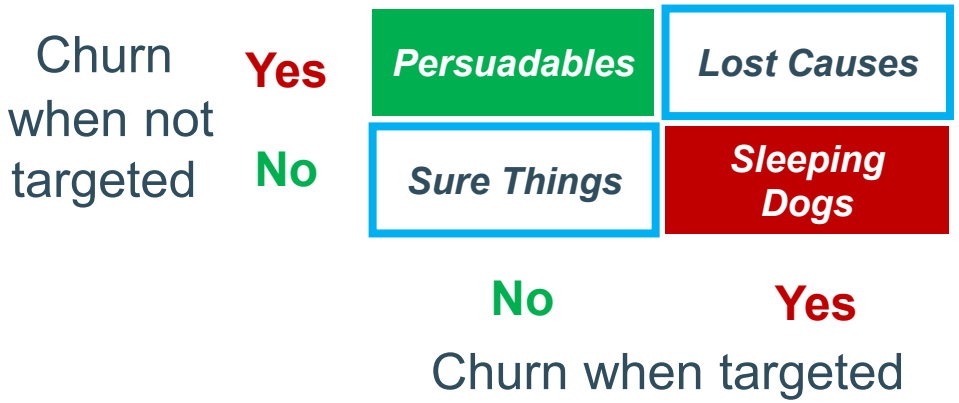
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Case: customer churn

Customers – Patients – Employees – Machines – …
Churn/response – Death – Turnover – Failure – …

Two issues with ***traditional*** predictive modeling solution design:

1. **Correct target variable?**



- **Instead of predicting what will happen …**

  … predicting what will happen depending on what we do

Case: customer churn

Factual: the outcome that occured

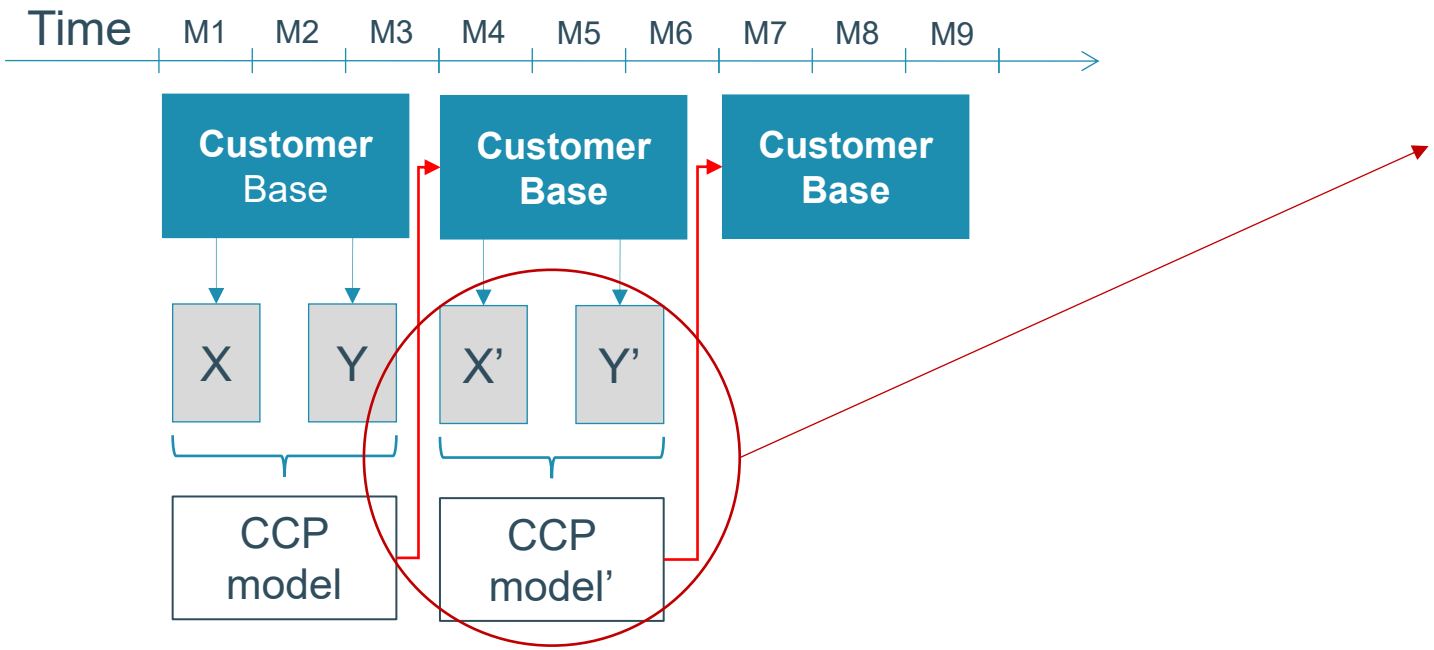
Counterfactual: the outcome that would have occurred if the other action had been applied (target or not)

Exercise: Monique?

Two issues with ***traditional*** predictive modeling solution design:

2. **Feedback loop(s): using a predictive model**

   … alters customer behavior!



X' Y' T

Customer	Age	Duration	Outcome	Targeted	Counter-factual
Pieter	41	5Y	churn	yes	churn
Eddy	55	24Y	churn	no	churn
Wouter	37	1Y	no churn	yes	churn
…	…	…	…	…	…
Monique	58	1Y	churn	yes	no churn

Y': recorded outcome in M6

- Y' = churn
  - Targeted? **Lost cause!**
  - Not targeted? **Unpredictable?**
- Y' = no churn
  - Targeted? **Persuadable!**

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Hidden Technical Debt in Machine Learning Systems

D. Sculley, Gary Holt, Daniel Golovin, Eugene Davydov, Todd Phillips
{dsculley,gholt,dgg,edavydov,toddphillips}@google.com
Google, Inc.

Dietmar Ebner, Vinay Chaudhary, Michael Young, Jean-François Crespo, Dan Dennison
{ebner,vchaudhary,mwyoung,jfcrespo,dennison}@google.com
Google, Inc.

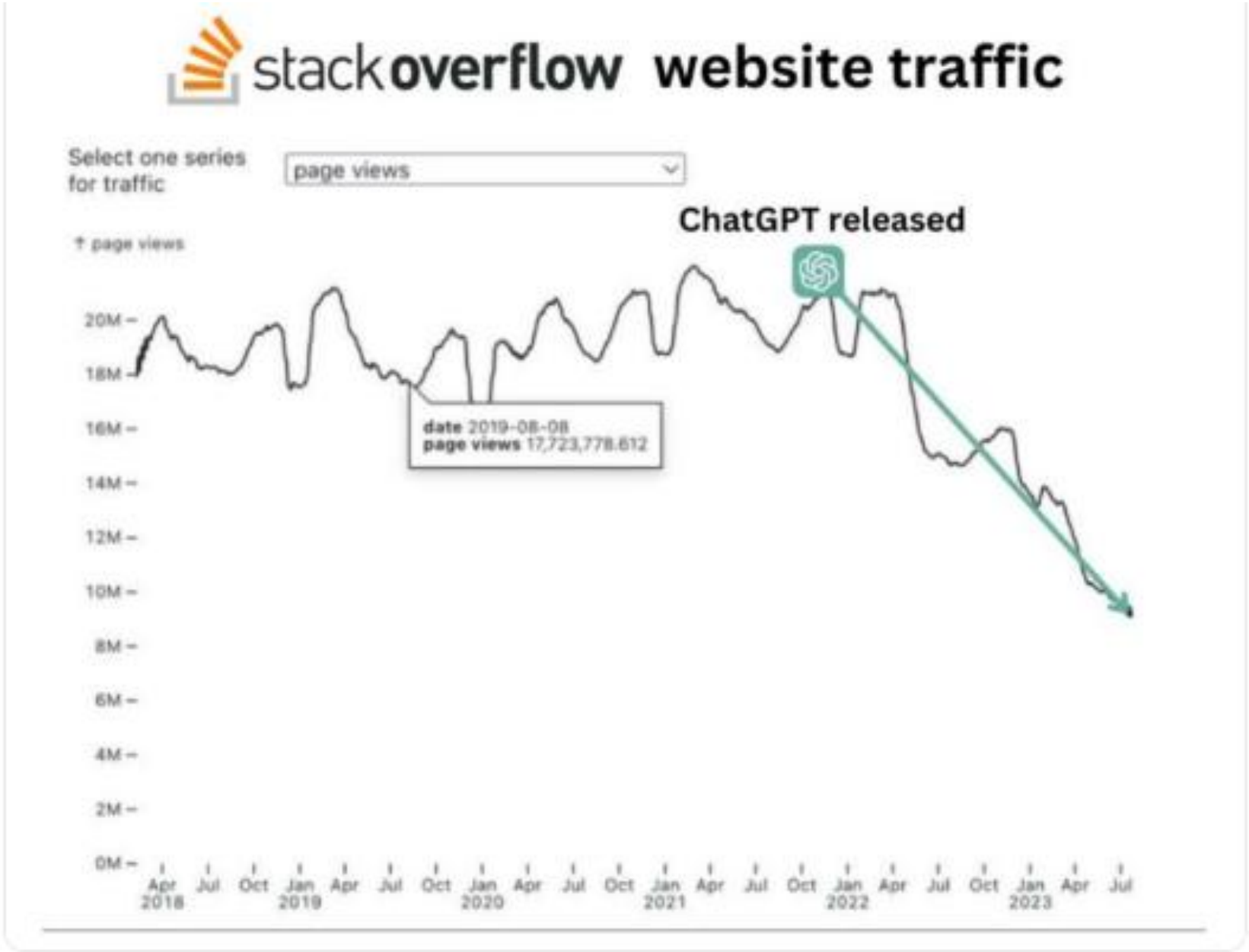
Abstract

Machine learning offers a fantastically powerful toolkit for building useful complex prediction systems quickly. This paper argues it is dangerous to think of these quick wins as coming for free. Using the software engineering framework of *technical debt*, we find it is common to incur massive ongoing maintenance costs in real-world ML systems. We explore several ML-specific risk factors to account for in system design. These include boundary erosion, entanglement, hidden feedback loops, undeclared consumers, data dependencies, configuration issues, changes in the external world, and a variety of system-level anti-patterns.

1. By using machine learning systems to impact the world …
2. … you do have an impact on the world …
3. … invalidating your machine learning system or creating problems for maintaining the system
   - Churn prediction
   - Credit scoring: rejects
   - …

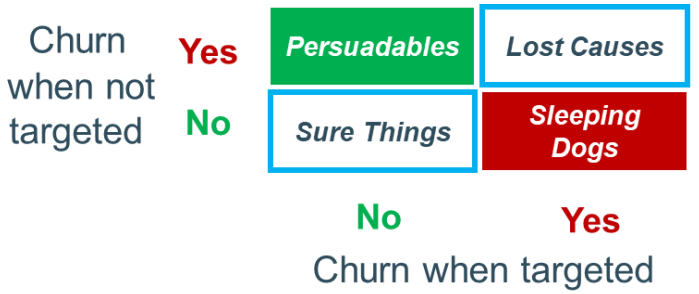
Other examples?

Sculley, D., Holt, G., Golovin, D., Davydov, E., Phillips, T., Ebner, D., ... & Dennison, D. (2015). Hidden technical debt in machine learning systems. *Advances in neural information processing systems*, *28*.


stackoverflow website traffic
Select one series for traffic
page views
ChatGPT released
↑ page views
20M
18M
16M
14M
12M
10M
8M
6M
4M
2M
0M
date 2019-08-08
page views 17,723,778,612
Apr 2018
Jul
Oct
Jan 2019
Apr
Jul
Oct
Jan 2020
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Jan 2021
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Case: customer churn



- **Solutions to issues?**
  - **Issue 1**: predict the outcome *conditional* on the intervention!
    - Instead of predicting what will happen …
      … we should predict what will happen **depending on what we do**
    - = predicting the **causal effect of the intervention**
  - **Issue 2**: incorporate information on previous interventions in the model!
    - Instead of predicting and ignoring the potential impact of the intervention …
      … we should predict **the outcome conditional on the intervention**
    - = predicting the **causal effect of the intervention**
  - **HOW?** **CAUSAL INFERENCE!**

2. Causal inference

What is causal inference?

What is causal machine learning?

What is uplift modeling

Causal inference

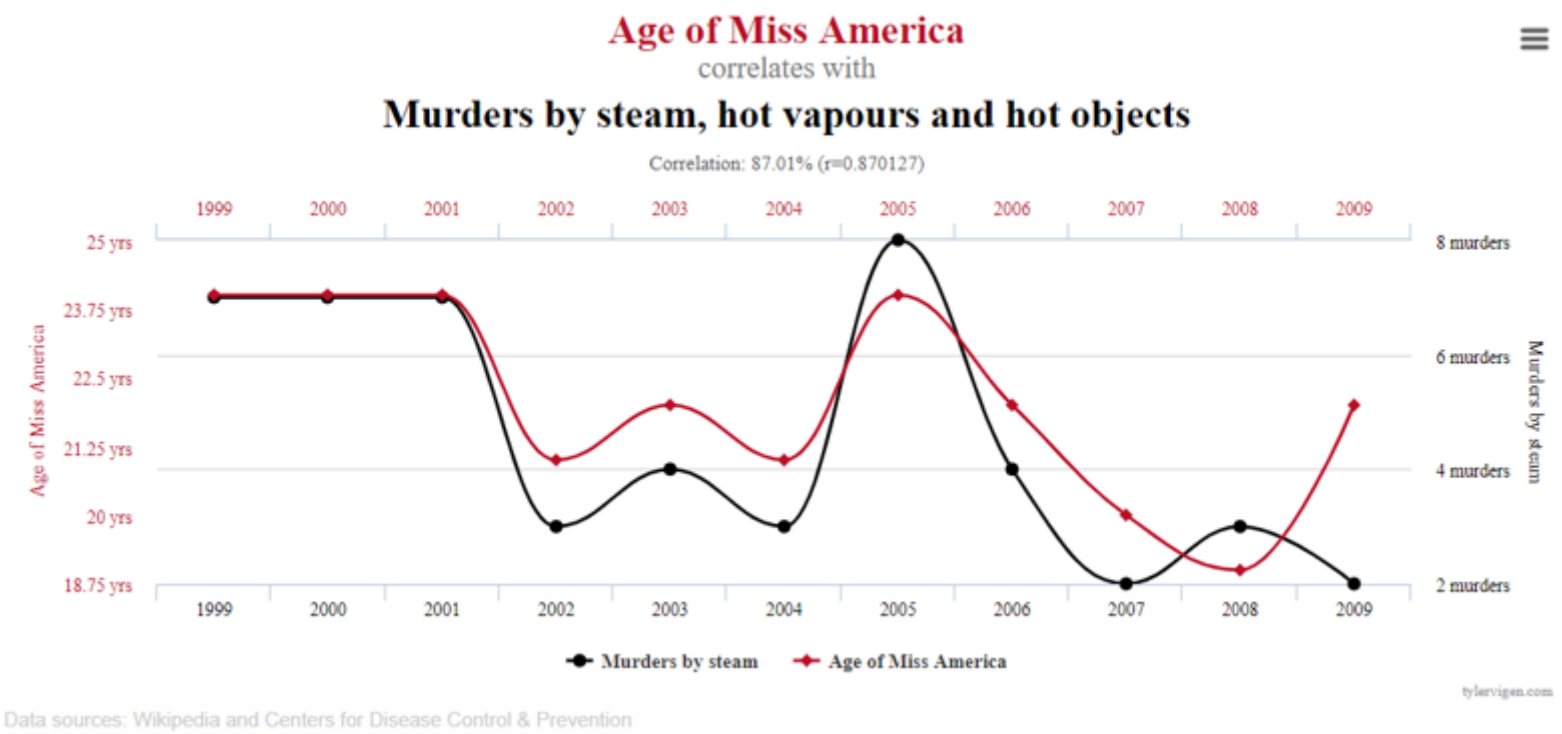
- Learning about cause and effect from data
- Challenging!

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Causal inference: challenges

Challenge: Spurious correlation

Also of importance in predictive analytics!

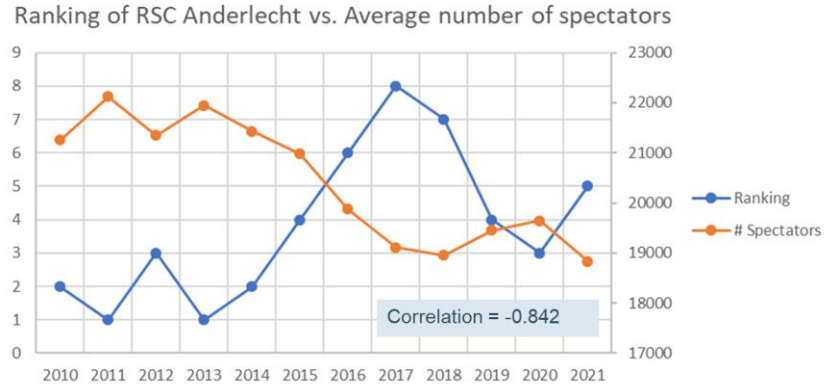


Causal effect?

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Causal inference: challenges

Challenge: Inversed causality



The case of RSC Anderlecht:

- RSC Anderlecht is a football team that has been underperforming for some years, whereas it used to top the Belgian league on a regular basis

- Analysis of historical data learns that, **if** there are more fans in the stadium, **then** there is a higher probability of RSC Anderlecht winning the game

Causal effect?

- So, to turn its dramatic performance over the past seasons, RSC Anderlecht calls upon its fans to come and watch the games, so as to achieve better results, since it is believed that more fans leads to better results

Causal inference: challenges

Challenge: Hidden confounder

The case of the student sleeping with his shoes on:

- Rudy was a student, enjoying a great time when studying in Leuven
- Rudy noticed that, **if** he slept with his shoes on, **then** he used to have a headache in the morning

Causal effect?

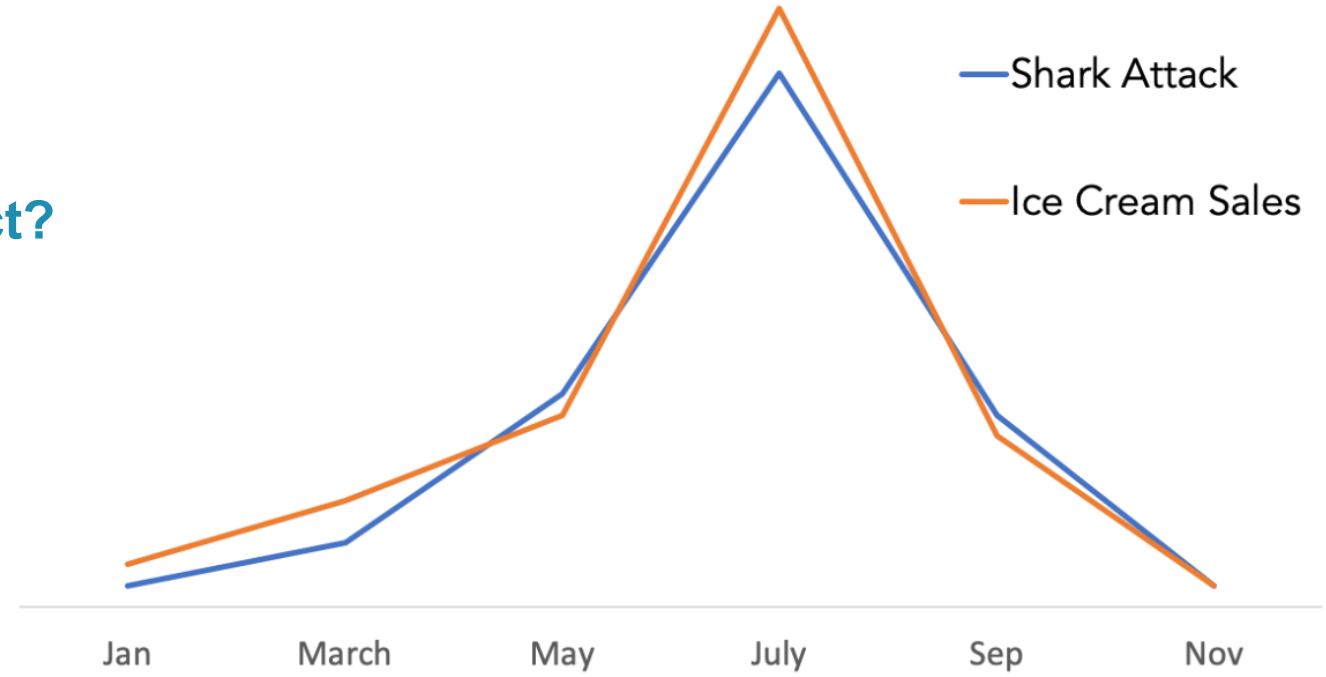
- So, he concluded not to sleep with his shoes on.
- He failed his course on statistics

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Causal inference: challenges

- Exercise

Causal effect?



Causal inference

Does causation imply correlation?

- Learning causal effects from data is challenging
  - **Correlation ≠ causation!**
  - But then, what if we want to estimate causal relations?

- In **machine learning**, this question popped up *rather recently*
- In **other fields**, researchers have been investigating causality since long:
  - Econometricians
  - Statisticians
  - …
- But: with somewhat **different objectives**!
  - There's causal inference … and there's causal inference

Causal inference

DAG vs. PO: Concerns a difference in ***WHAT***

Causal machine learning = causal inference with ML
Concerns a difference in ***HOW***

~ Statistics vs. Machine Learning

- **Causal inference**: two ***schools***

Pearl

Causal graph learning: '***to explain***'

- Causal **discovery** and **effect** learning
- Cause is not necessarily *controllable*
- E.g.: learning about events that trigger churn
- **Directed Acyclic Graph (DAG) framework**

Rubin (& Neyman & Imbens)

Causal effect learning: ***'to predict'***

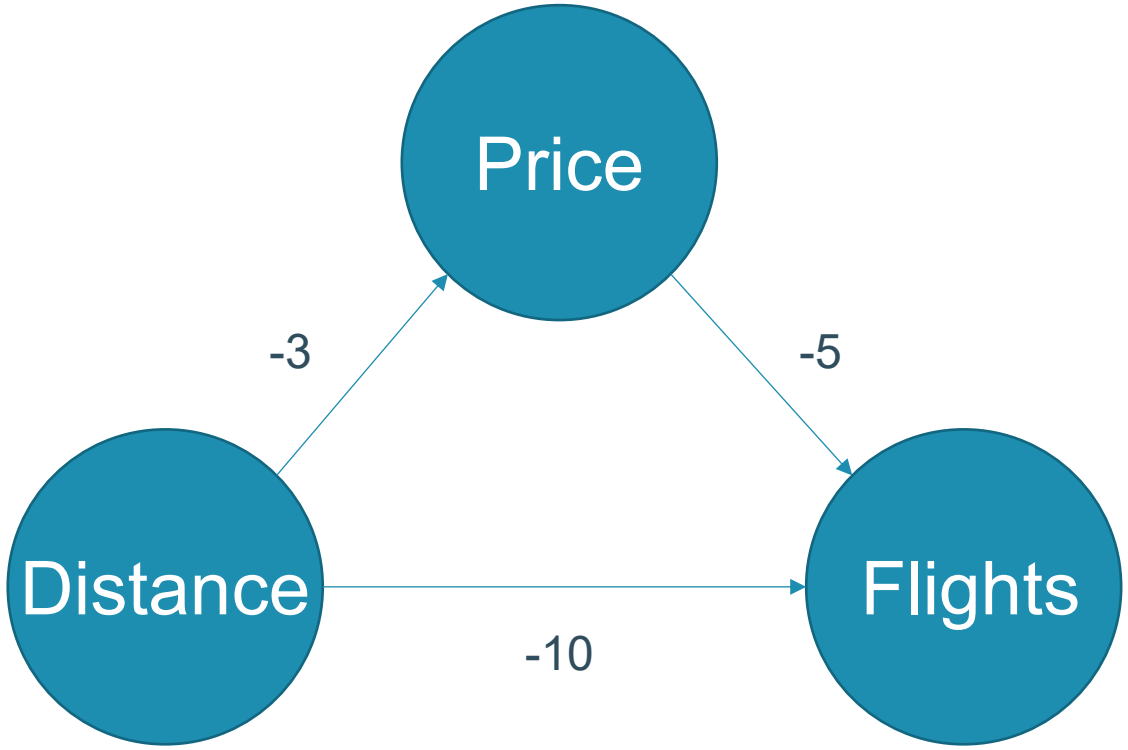
- Model the causal **effect** of a **treatment** on an outcome of interest
- Typically, a treatment that we *control*
- **Potential Outcome (PO) framework**

ITE / CATE / HTE / Uplift

Shmueli, G. (2010). To explain or to predict?. *Statistical science*, *25*(3), 289-310.

Causal inference: DAG

Conclude from model: maximize distance?



Causal graph – airport case

Effect of distance on # Flights?

- **Direct effect** (*Distance → Flights)*: Negative perception of customer to distance
- **Indirect effect** (*Distance → Price → Flights)*: Increase in distance, forces companies to reduce the price and that produces an increment of flights

$$Flights := -10 \cdot Distance - 5 \cdot Price + \varepsilon$$

$$Price := -3 \cdot Distance + \varepsilon$$

Structural Equation Models (SEM)

$$Flights := +5 \cdot Distance + \varepsilon$$

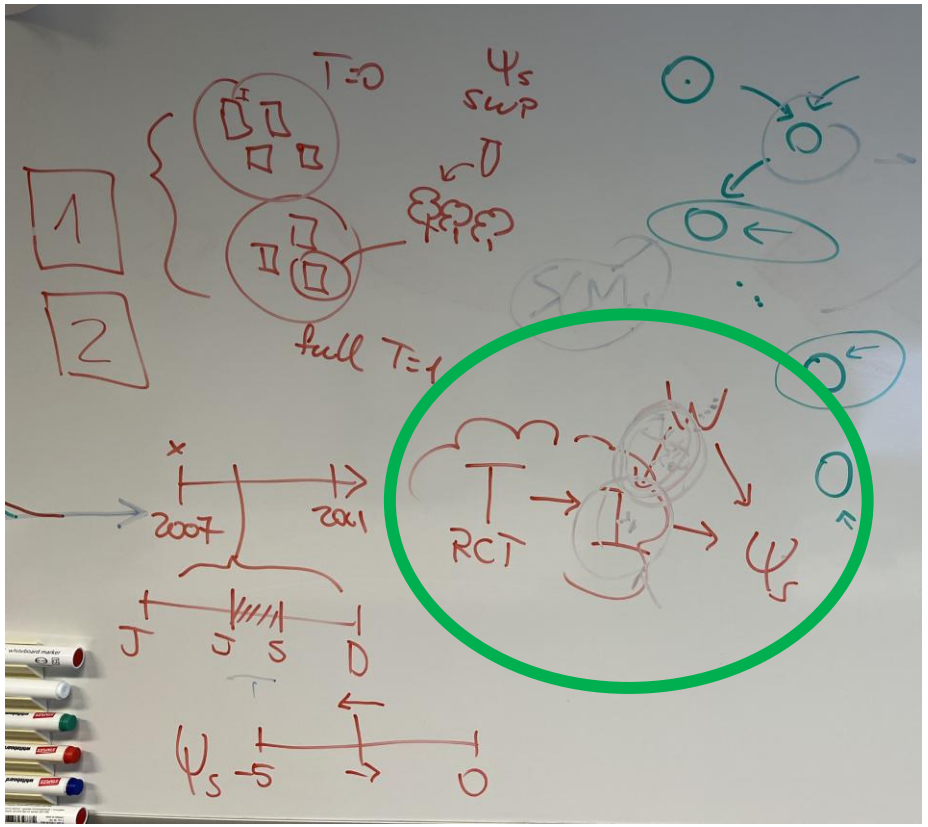
https://towardsdatascience.com/use-causal-graphs-4e3af630cf64

Causal inference: DAG

Data gathered in an experimental setup to learn about the impact of irrigation on yield and water stress because of drought in orange trees

In order to **optimize the amount of irrigation**, one may consider to attempt learning the **causal effect of irrigation**

ψ_s: stemwater
I: # irrigation
T: decision to irrigate in RCT
W: weather



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Causal inference: PO

Case: customer churn → business objective:

- Optimal allocation of treatments (retention incentives)
- To maximize positive outcome rate (retention rate) **= goal of uplift modeling**

How? → Predict-then-optimize approach

- **Step 1**:
  - **Predict**
  - At the **individual** instance level
  - The net effect
  - Of applying the treatment
  - On the outcome of interest

Net effect of a treatment on the outcome of interest at the individual instance level

= Individual Treatment Effect (ITE)

- **Step 2**:
  - **Optimize** selection of instances to treat based on predictions to maximize positive outcome rate

Causal inference: PO

Case: customer churn → business objective:

- Optimal allocation of treatments (retention incentives)
- To maximize positive outcome rate (retention rate) **= goal of uplift modeling**

How? → Predict-then-optimize approach



Contents lists available at ScienceDirect

European Journal of Operational Research

journal homepage: www.elsevier.com/locate/eor

Analytics, Computational Intelligence and Information Management

Uplift modeling with continuous treatments: A predict-then-optimize approach

Simon De Vos [a,*], Christopher Bockel-Rickermann [a], Stefan Lessmann [b,c], Wouter Verbeke [a]

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[c] Bucharest University of Economic Studies, 6 Piata Romana, Bucharest, 010374, Romania

ABSTRACT

The goal of uplift modeling is to recommend actions that optimize specific outcomes by determining which entities should receive treatment. One common approach involves two steps: first, an inference step that estimates conditional average treatment effects (CATEs), and second, an optimization step that ranks entities based on their CATE values and assigns treatment to the top k within a given budget. While uplift modeling typically focuses on binary treatments, many real-world applications are characterized by continuous-valued treatments, i.e., a treatment dose. This paper presents a predict-then-optimize framework for uplift modeling with continuous treatments. First, in the inference step, conditional average dose responses (CADRs) are estimated from data using causal machine learning techniques. Second, in the optimization step, we frame the assignment task of continuous treatments as a dose-allocation problem and solve it using integer linear programming (ILP). This approach allows decision-makers to efficiently and effectively allocate treatment doses while balancing resource availability, with the possibility of adding extra constraints like fairness considerations or adapting the objective function to take into account instance-dependent costs and benefits to maximize utility. The experiments compare several CADR estimators and illustrate the trade-offs between policy value and fairness, as well as the impact of an adapted objective function. This demonstrates the framework's advantages and flexibility across diverse applications in healthcare, lending, and human resource management. All code is available on https://github.com/SimonDeVos/UMCT.

De Vos, S., Bockel-Rickermann, C., Lessmann, S., & Verbeke, W. (2025). Uplift modeling with continuous treatments: A predict-then-optimize approach. *European Journal of Operational Research*.

Causal inference: PO

Can you think of extensions to this base case?
Refinements in specifying the business objective?

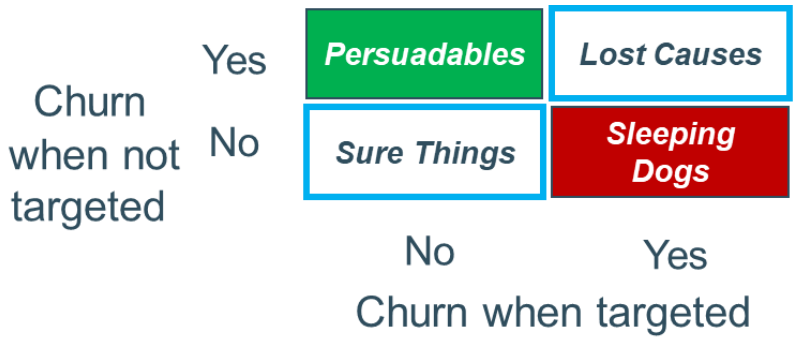
Base case:

- Binary **treatment** (i.e., decision or action)

  → Binary treatment variable (T)

- Binary **outcome**

  → Binary target variable (Y)



Business objective:

- Optimal allocation of treatments
- To maximize positive outcome rate

Causal inference: PO

Base case extensions:

- Binary **treatment** → multiclass, continuous, sequential, time-dependent treatments, …
- Binary **outcome** → multiclass, continuous, multiple, time-dependent outcomes,…

Business objective extensions:

- Optimal allocation of treatments

→ Constraints on number or size of treatments (e.g., limited capacity, budget, etc.)

- To maximize positive outcome rate

→ To maximize **returns** (with variable costs & benefits of instances & treatments)

Causal inference: PO

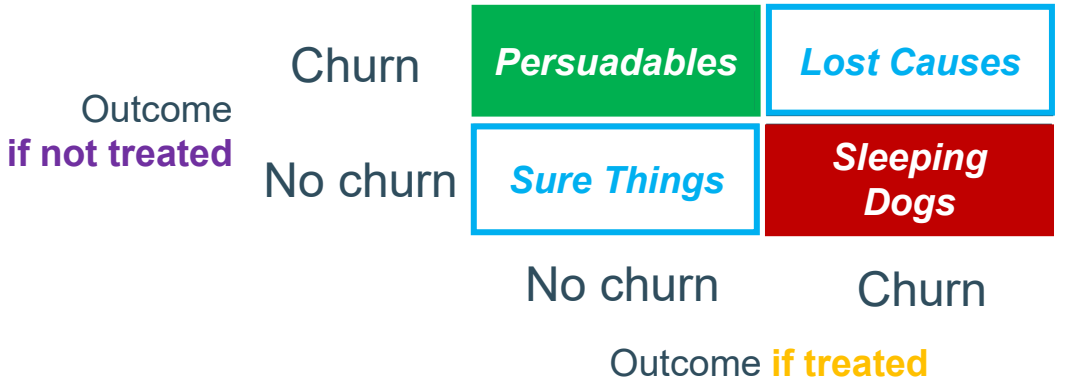
Counterfactual: the outcome that would have occurred if the other action had been applied (target or not)

Potential outcome: the outcome that will happen for a potential action that can be applied

- ITE = Individual Treatment Effect
  - Difference between the **potential outcomes**

$$ITE = \underbrace{Y(\boldsymbol{T=1})}_{\text{Outcome if treated}} - \underbrace{Y(\boldsymbol{T=0})}_{\text{Outcome if not treated}}$$

- E.g., net effect of retention offer on churn outcome:
  - $Y = 1$: Churn
  - $Y = 0$: No churn
  - ITE = -1 : **persuadables**
  - ITE = 0 : **lost causes or sure things**
  - ITE = 1 : **sleeping dogs**



- Conditional Average Treatment Effect:

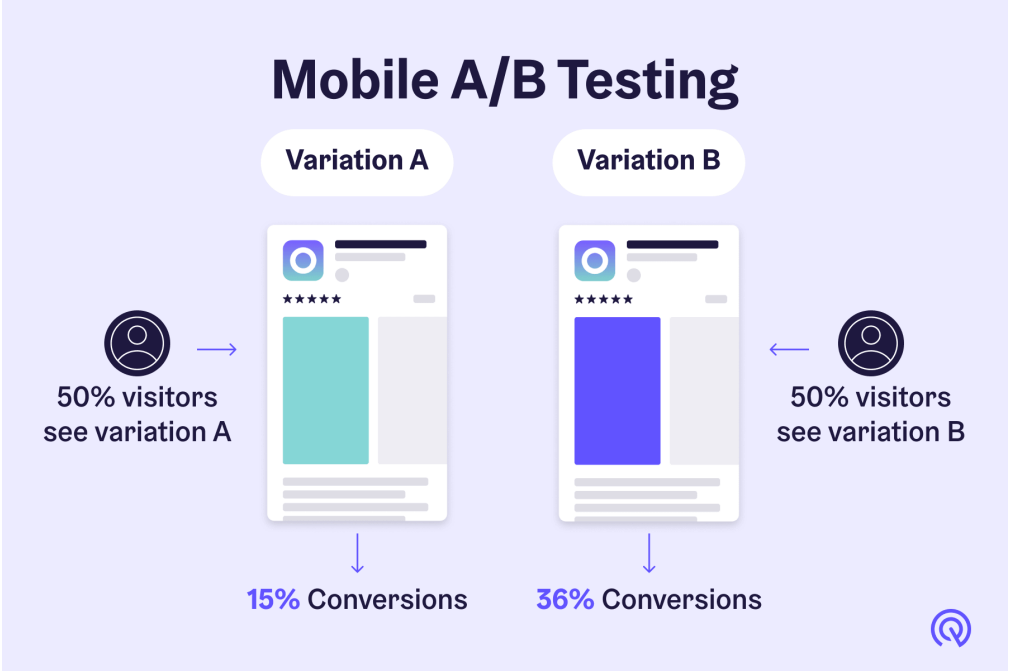
$$CATE = \overbrace{P(Y=1|X,\boldsymbol{T=1})}^{\text{Churn probability if treated}} - \overbrace{P(Y=1|X,\boldsymbol{T=0})}^{\text{Churn probability if not treated}}$$

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Causal inference: PO

- ITE = Individual Treatment Effect
- **ATE = Average Treatment Effect**
  - Average
  - Net effect
  - On the outcome of interest
  - Of applying the treatment
  - Across the population

- **Examples (of the use of) ATE?**
- Establishing the ATE:
  - A/B tests, clinical tests, …
  - Randomized Controlled Trials (RCT)



https://appradar.com/academy/app-store-listing-a-b-testing

3. Challenges

.. in learning about **cause and effect** from data

(both in the DAG and PO framework)

Challenges

- Correlation does not imply causation!
  - Spurious correlations
  - Inversed causality

  → **Not (really) of importance in treatment effect estimation**
  → **Important in DAG framework**

  - (Hidden) confounders

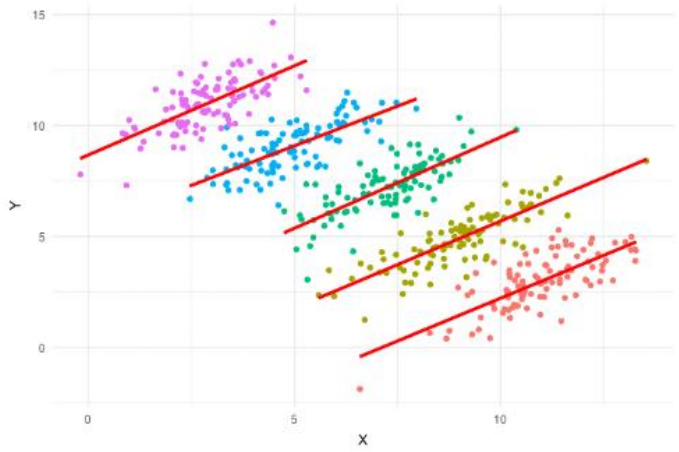
  → **Of particular importance in treatment effect estimation**

- Important? Wrong decisions!
  - Call on fans to come and watch
  - Not sleeping with shoes on

Challenges

Challenge: *unhide* hidden confounders!

Simpson's paradox: ATE inverse of CATE



- Case: Machine maintenance
  - X: # maintenance
  - Y: # time between failures
  - Hidden confounder! (Any others?)
- **Exercise**: hidden confounders?
  - Post-operative pain vs. patient satisfaction @UZ
  - Mortgage pricing – acceptance risk modeling

- **Conclusion**: (hidden) confounders lead to both:
  1. Wrong conclusions w.r.t. causal relationships in **causal discovery**
     - E.g., shark attacks & ice cream sales
  2. Wrong models and estimates of treatment effects in **causal effect learning**

Challenges

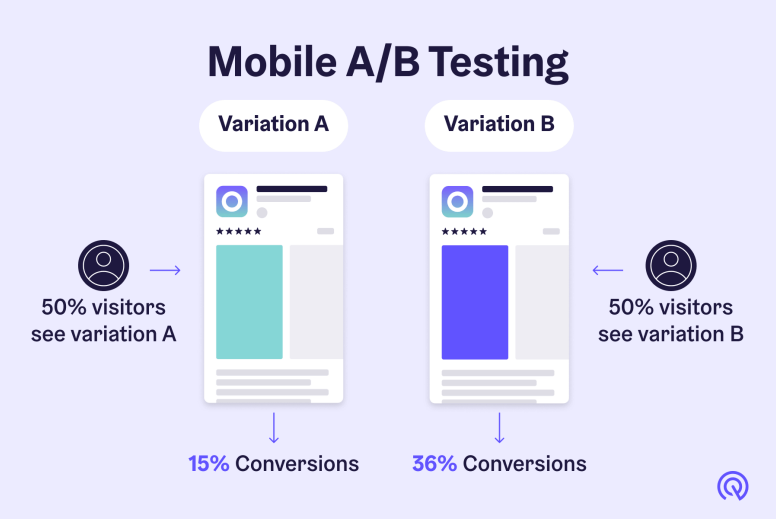
Challenge: obtaining RCT data!

Challenge: RCT for continuous treatment?

→ Experimental design + modeling

Solution (for treatment effect learning, not for causal discovery)

- Control group vs. treatment group: **randomized controlled trials (RCT)**
- '…*all else equal*…': to isolate the (average) effect of the treatment from confounders

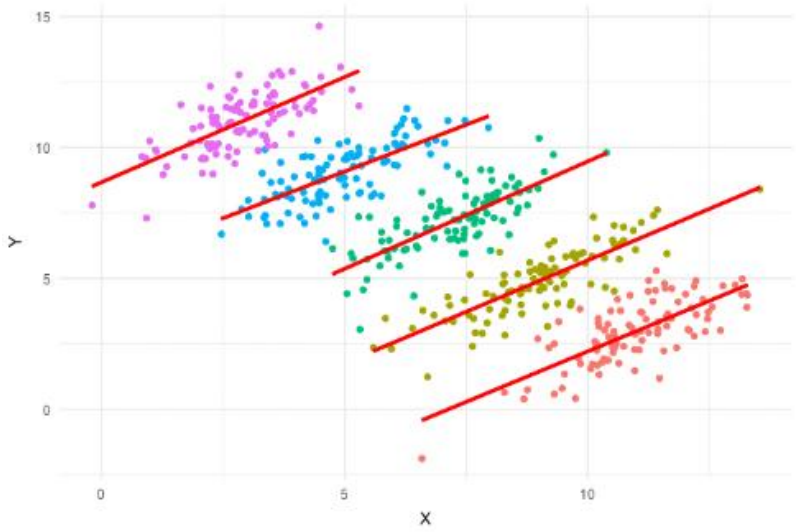


Traditional scientific approach for learning about cause and effect:

- ATE
  - A/B testing
  - Clinical trials
- ITE
  - Uplift modeling
  - Personalized medicine

Challenges

- Case: Machine maintenance
- Note that in this data there are no observations of:
  - **Old machines** receiving **little maintenance**
  - **New machines** receiving a **lot of maintenance**



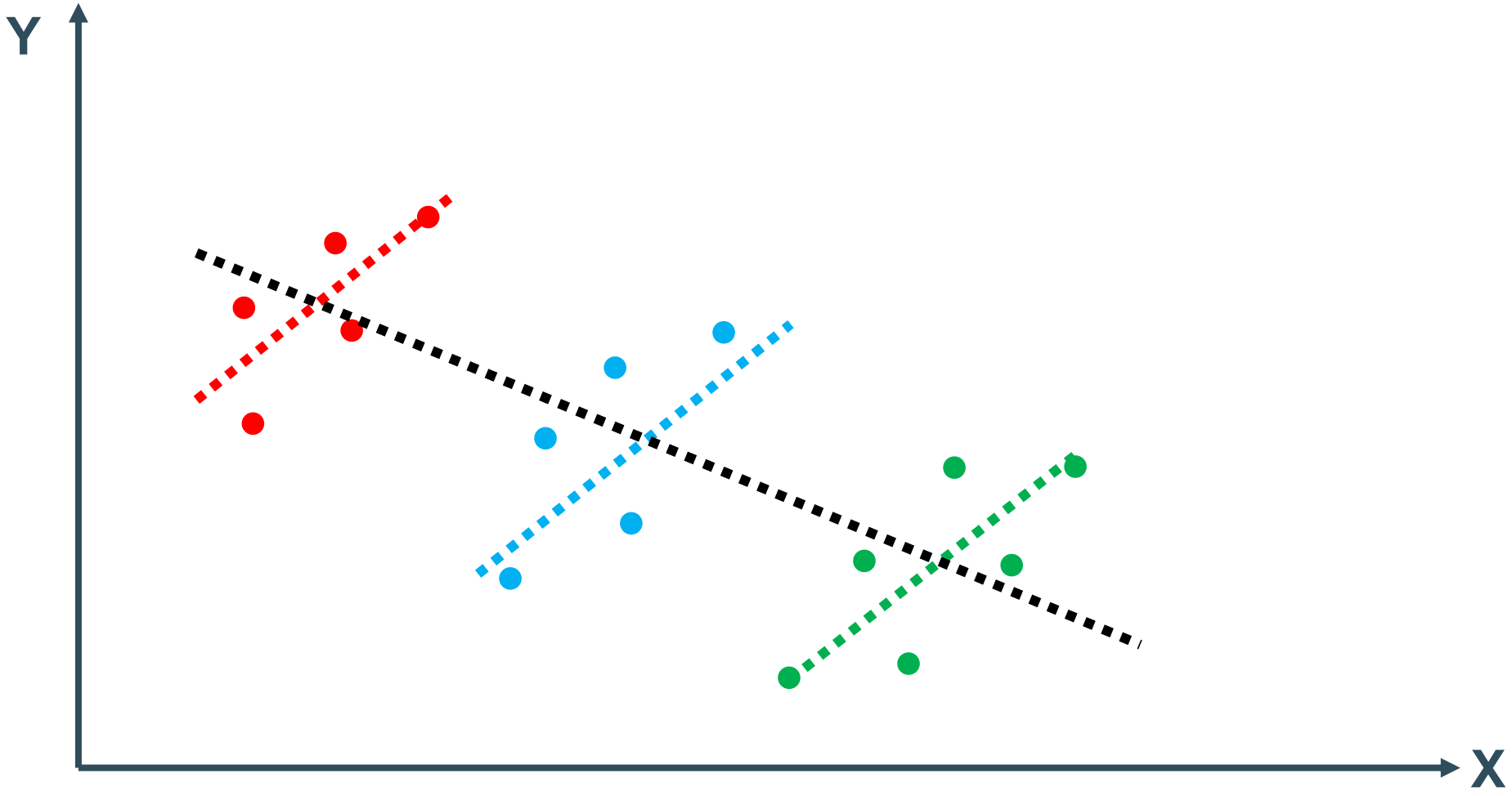
This is observational data!

- Learning from observational data?
  - Be cautious for **assignment bias** in model learned from observational data!
  - Information used by the assignment policy should be included in the data!

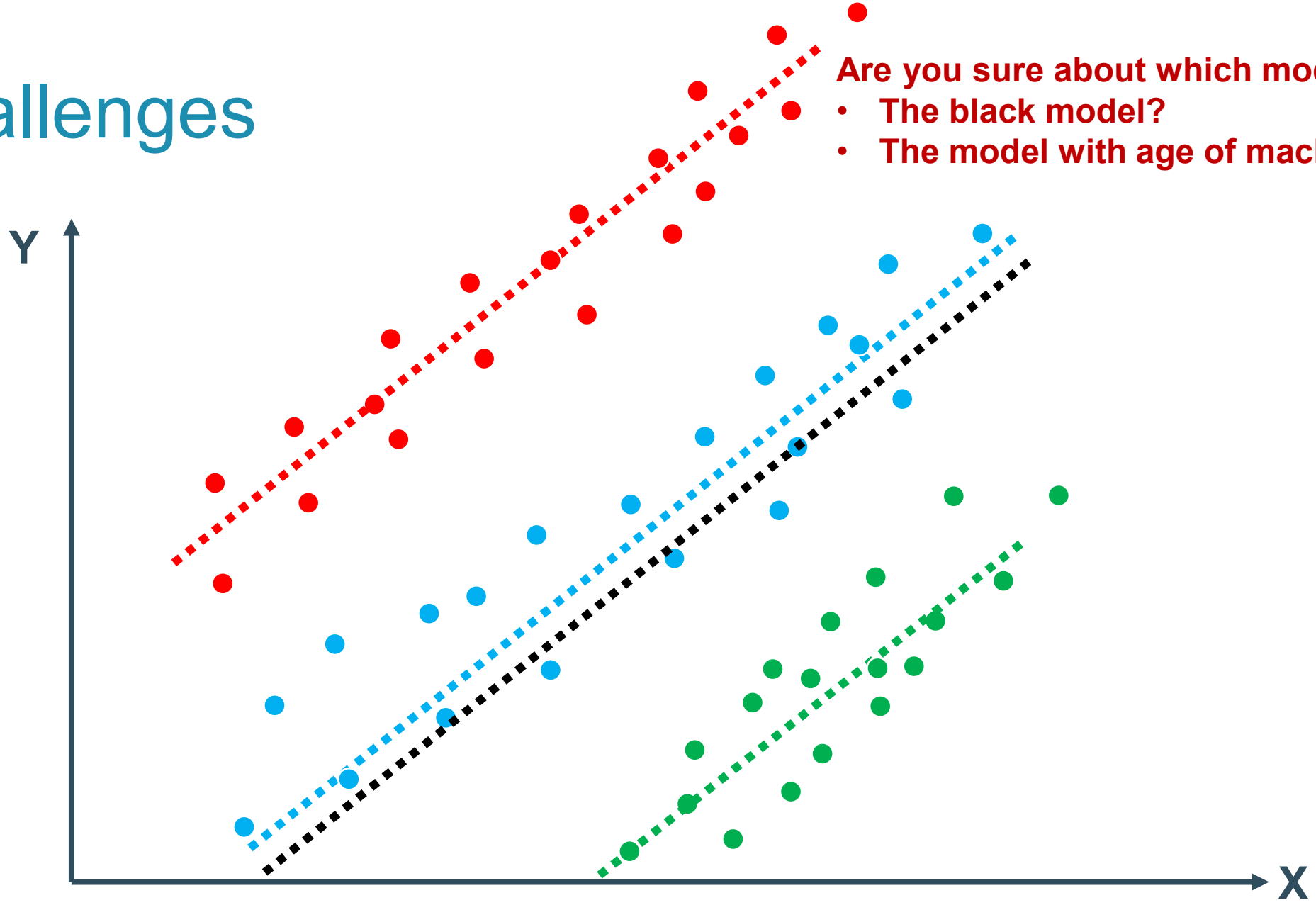
Challenges

Are you sure about which model is *correct*?

- **The black model?**
- **The model with age of machines included?**



Challenges



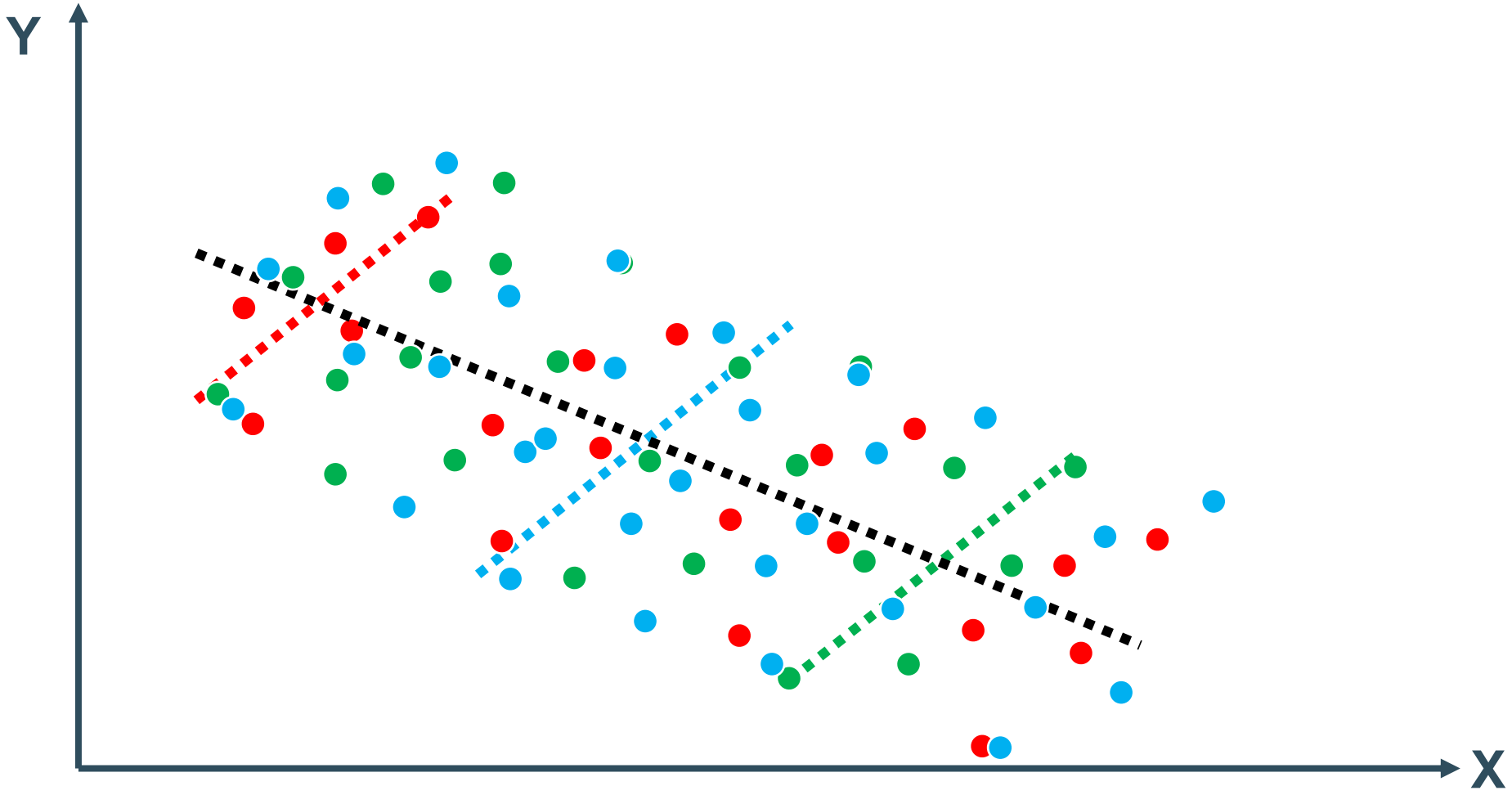
Are you sure about which model is *correct*?

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Challenges

Are you sure about which model is *correct*?

- **The black model?**
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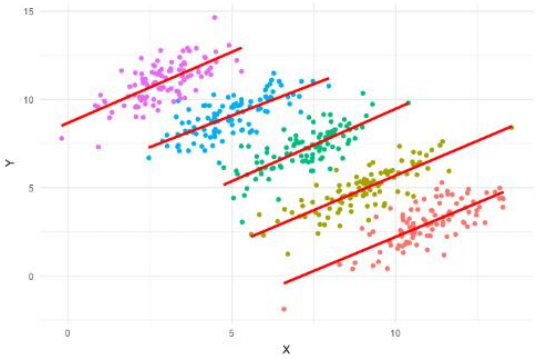


Challenges

Challenge: learning from observational data!

- **What if we cannot obtain RCT data?**
  - Case: Machine maintenance
  - Case: OrganITE

No observations of:
- Old machines receiving little maintenance
- New machines receiving a lot of maintenance



Observational data → Confounding bias!

- Can we learn from observational data?
  - **Ignorability assumption**: $(Y_{T=0}, Y_{T=1}) \perp T|X$
    - A.k.a. conditional independence assumption
    - No hidden confounders: **treatment assignment drivers** as covariates in the data set
  - **Overlap assumption:** $\delta < P(T|X) < 1 - \delta \quad \forall (X,T) \quad and\ \delta \in [0,1)$
    - A.k.a., common support assumption
    - Every entity in the population can be in each group, although possibly not as likely
  - **SUTVA**, **consistency**, …

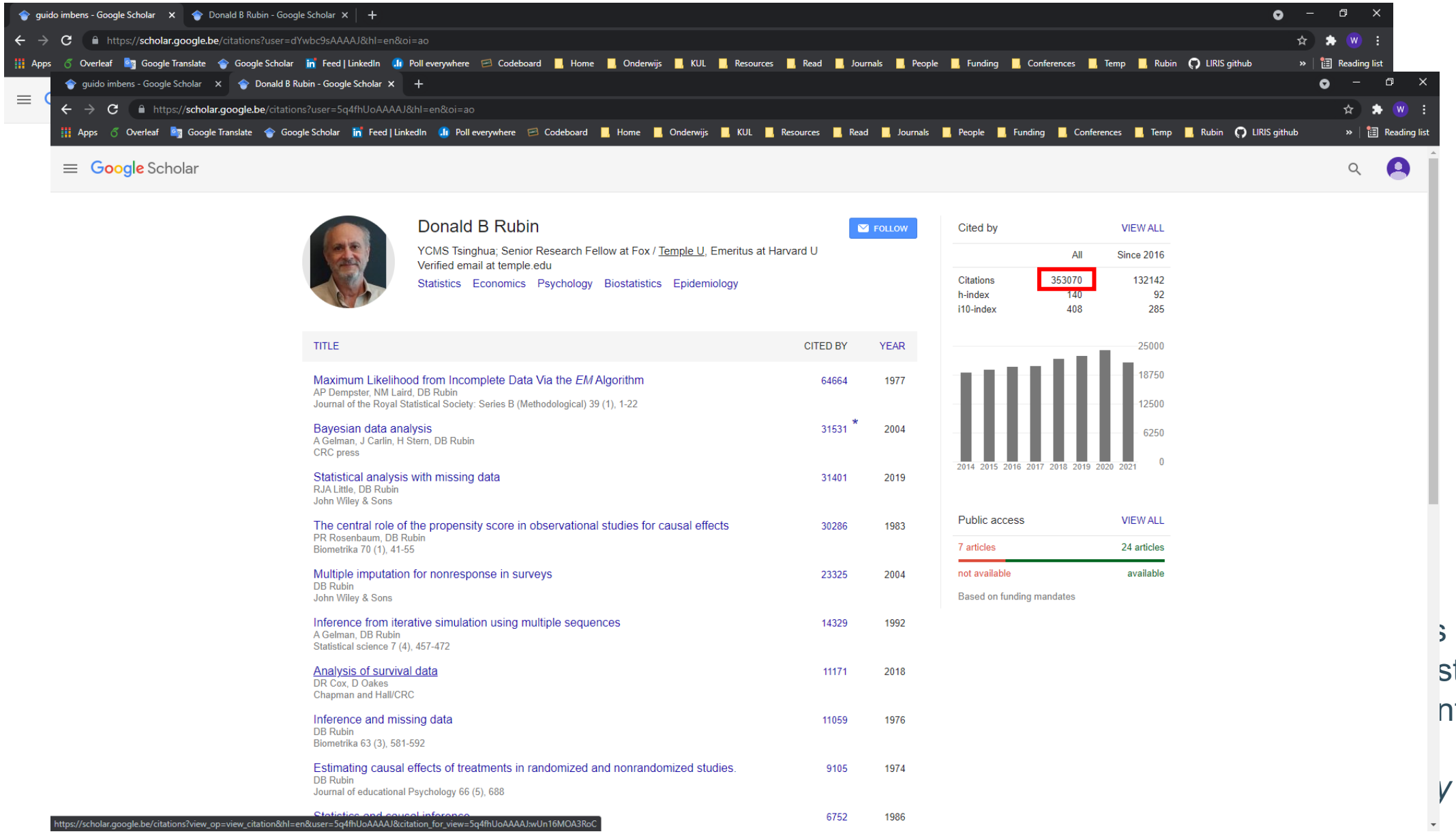
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Challenges

Learning from “natural experiments”?



Different countries implement different policies, staging “natural experiments”



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Challenges

Challenge: learning to predict an unobserved target variable!

- You cannot simultaneously
  - Apply a treatment to an individual
  - **And** not apply a treatment to that individual
- So, **you cannot measure the difference in outcome at the individual level**
  - E.g., to observe whether an individual is a persuadable, a lost cause,…

- We do not observe the individual treatment effect!
  - I.e., the target variable Y that we want to predict!
  - Supervised learning?

Fundamental Problem of Causal Inference

Challenges

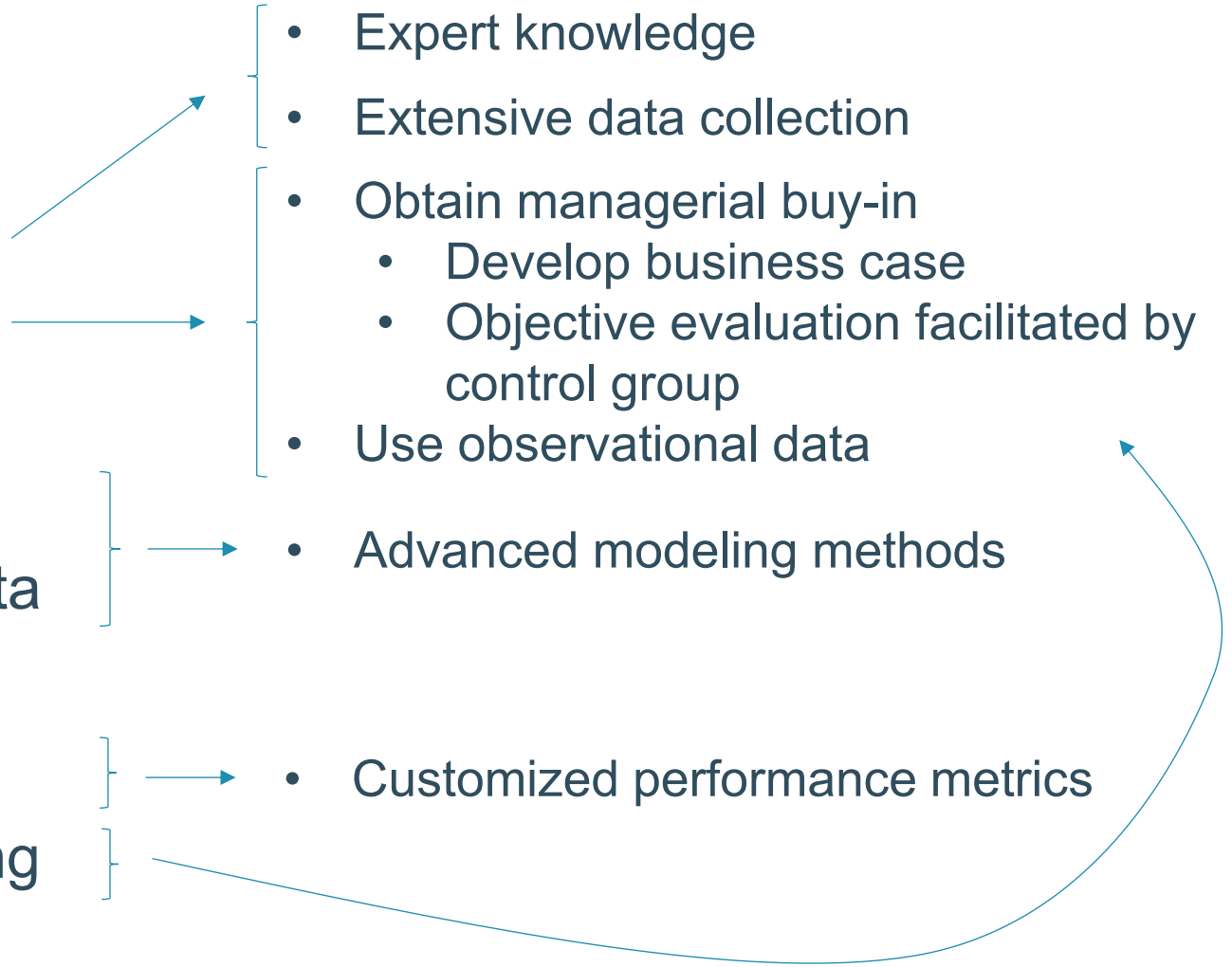
Challenge: evaluating estimates for an unobserved target variable!

- Suppose we developed an ITE model:
  - To assess the performance
  - We need to compare the predicted ITE with the observed ITE
  - But:
    - We cannot observe the ITE
    - The applied treatment may be different from the optimal treatment
  - So, **how to use historical data to evaluate an ITE model**?

- Once the model is implemented, how can we **monitor model performance**?

Challenges: summary

- Pre-processing:
  - Identifying hidden confounders
  - Data requirements for learning
- Processing:
  - Non-observed target variable
  - Learning from observational data
- Post-processing:
  - Non-observed target variable
  - Data requirements for monitoring



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Chapter 9. Prescriptive Analytics: Causal Machine Learning

Data Science for Business

D0I74A

Prof. dr. Wouter Verbeke

Prof. dr. Bart Baesens



Course schedule

BLOCK2			
6	26/10/2026	Chapter 6. Predictive Analytics - Artificial Neural Networks	
6	27/10/2026	**No class~~, video-lecture: Chapter 7. Predictive Analytics - Forecasting~~**	
7	02/11/2026	**No class, university holiday**	
7	03/11/2026	Chapter 9. Prescriptive Analytics - Causal Machine Learning	
8	09/11/2026	**No class~~, video-lecture: Chapter 8. Descriptive Analytics~~**	
8	10/11/2026	Chapter 9. Prescriptive Analytics - Causal Machine Learning	**Report 1 deadline**
9	16/11/2026	Chapter 10. Prescriptive Analytics - Profit-driven Analytics	
9	17/11/2026	Guest Lecture #1: Toon Vanderschueren, Ph.D., Causal ML @Telenet	**Presence expected**
	17/11/2026	Assignment 2	**Report 2 > 14/12**
10	23/11/2026	Chapter 11. Prescriptive Analytics - Multi-armed Bandits & Reinforcement Learning	
10	24/11/2026	Guest Lecture #2: Dr. Véronique Van Vlasselaer, Ph.D., Business Analytics @SAS	**Presence expected**
BLOCK3			
11	30/11/2026	Chapter 12. Natural Language Processing	
11	01/12/2026	Chapter 12. Natural Language Processing	
12	07/12/2026	Chapter 13. Large Language Models & AI Agents	
12	08/12/2026	Chapter 13. Large Language Models & AI Agents	
13	14/12/2026	Feedback Assignment 2 Sample Exam Questions	**Report 2 deadline**
13	15/12/2026	Guest Lecture #3: To be confirmed	**Presence expected**

Outline

1. Case: customer churn
2. Causal inference
3. Challenges
4. Methods
5. Evaluation
6. Cases
7. Concluding remarks

4. Methods

3.1 Meta-learners

3.2 Deep learning

3.3 Deep learning with balancing

Challenges: summary

- Pre-processing:
  - Identifying hidden confounders
  - Data requirements for learning
- Processing:
  - Non-observed target variable
  - Learning from observational data

  → Advanced modeling methods
- Post-processing:
  - Non-observed target variable
  - Data requirements for monitoring

3.1 Meta-learners

3.2 Artificial Neural Networks

3.3 ANN with balancing

Where did we encounter meta-learners before?

4.1 Meta-learners

Meta-learners

- Modeling **strategy** or **framework** to estimate the **conditional average treatment effect (CATE)** that can be implemented with **any ML method**
  - Base learner
  - Cf. ensemble methods
- Different meta-learners:
  - **T-learner**
  - **S-learner**
  - X-learner
  - R-learner
  - DR-learner
  - …
  - **Transformed outcome method**
- Appropriate learner? Depends on the data generating process!

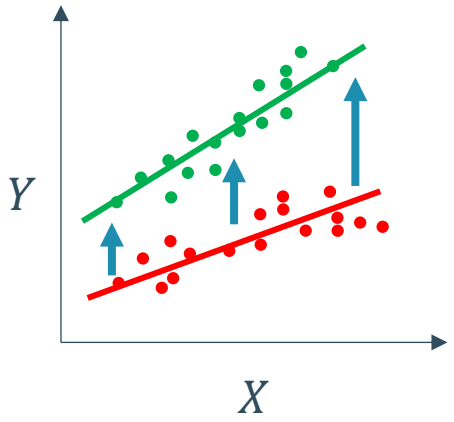
T-learner

Note: continuous outcome, binary treatment case

- **Two-model learner**

Treatment group

Control group



$$Y^{T=1} = \beta_0^{T=1} + \beta_1^{T=1} X$$

$$Y^{T=0} = \beta_0^{T=0} + \beta_1^{T=0} X$$

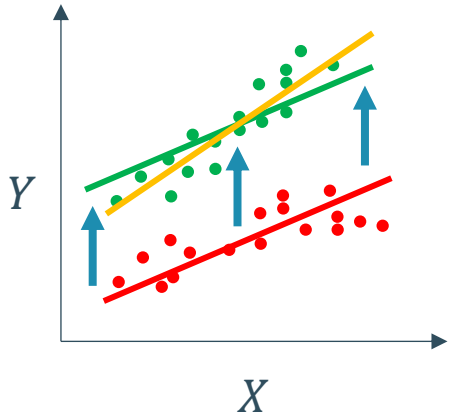
$$\hat{\tau}(x) = \beta_0^{T=1} + \beta_1^{T=1} X - \beta_0^{T=0} + \beta_1^{T=0} X$$

S – learner

- **S**ingle-model **learner**
  - Limited flexibility?
  - Add **interaction effects** to extend model flexibility (ATE > CATE)

Treatment group

Control group



$$Y = \beta_0 + \beta_1 X + \beta_2 T + \beta_3 XT$$

β_2 is the estimated average treatment effect (ATE)

S-learner

- **S**ingle-model **learner**

Customer	Age	Income	…	Treatment t	Target y
John	32	1,530	…	1	1
Sophie	48	2,680	…	1	0
…	…	…	…	…	…
Josephine	23	1,720	…	0	0
Bart	39	2,390	…	0	1
…	…	…	…	…	…

Pearl's do-operator: **P(Y|X,do(T))**

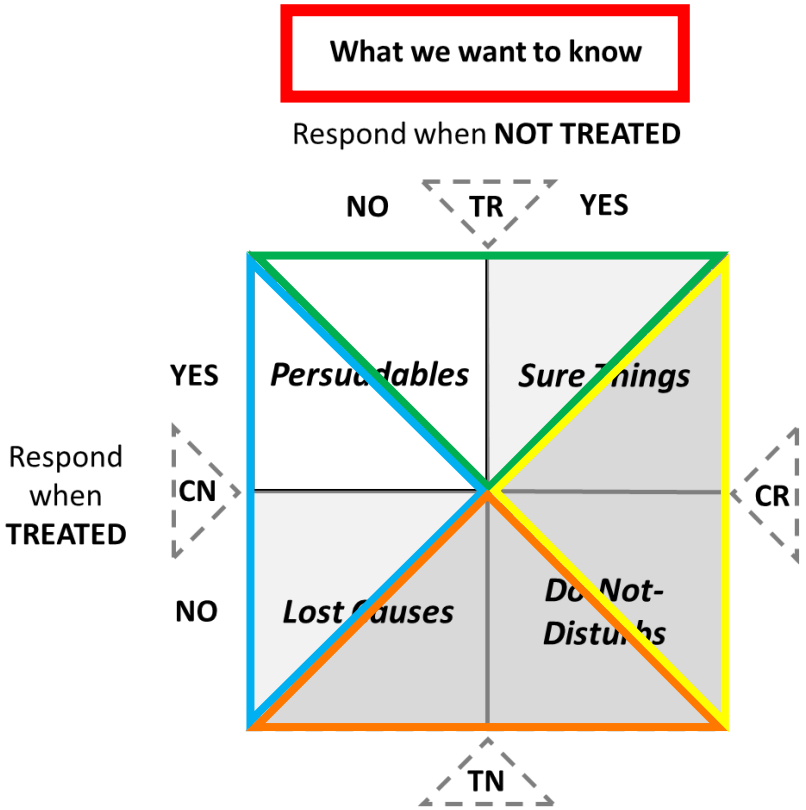
$$p(y = 1|x_1, \dots, x_k, \boldsymbol{t}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \underbrace{\beta_{k+1} \boldsymbol{x_1 t} + \cdots + \beta_{k+k} \boldsymbol{x_k t}}_{\text{Interaction effects}} + \underbrace{\beta_{k+k+1} \boldsymbol{t}}_{\text{ATE}})}}$$

- Model becomes a **simulation model**!
- Can be applied with any predictive modeling technique (RF, ANN, …)
- A.k.a. treatment dummy approach

Transformed outcome method

Response = no churn
No response = churn

Depending on the **two potential outcomes**



Depending on the **applied treatment** and the **observed outcome**

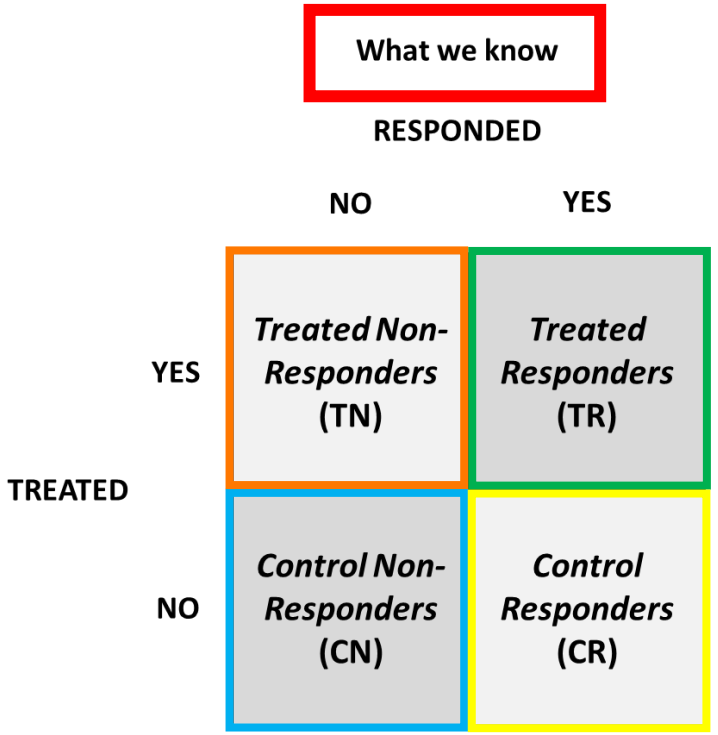
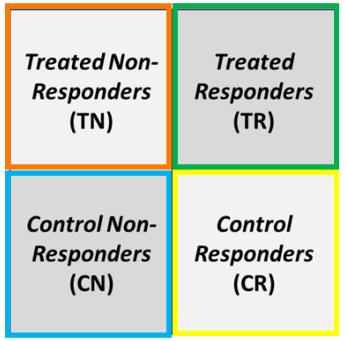
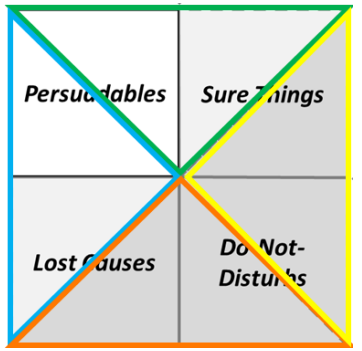


Figure adopted from Kane, 2014

Transformed outcome method





- Transformation of the outcome (y → y'):
  - Persuadables are treated responders or control non-responder: **positive class**
    - Treated responder: t = 1 and y = 1 → y' = 1
    - Control non-responder: t = 0 and y = 0 → y' = 1
  - Sleeping dogs are treated non-responders or control responders: **negative class**
    - Treated non-responder: t = 1 and y = 0 → y' = 0
    - Control responder: t = 0 and y = 1 → y' = 0

x_1	x_2	...	t	y
32	1,530€	...	1	1
48	2,680€	...	1	0
...	...	...	...	...
23	1,720€	...	0	0
39	2,390€	...	0	1
...	...	...	...	...

⇨

y'
1
0
...
1
0
...

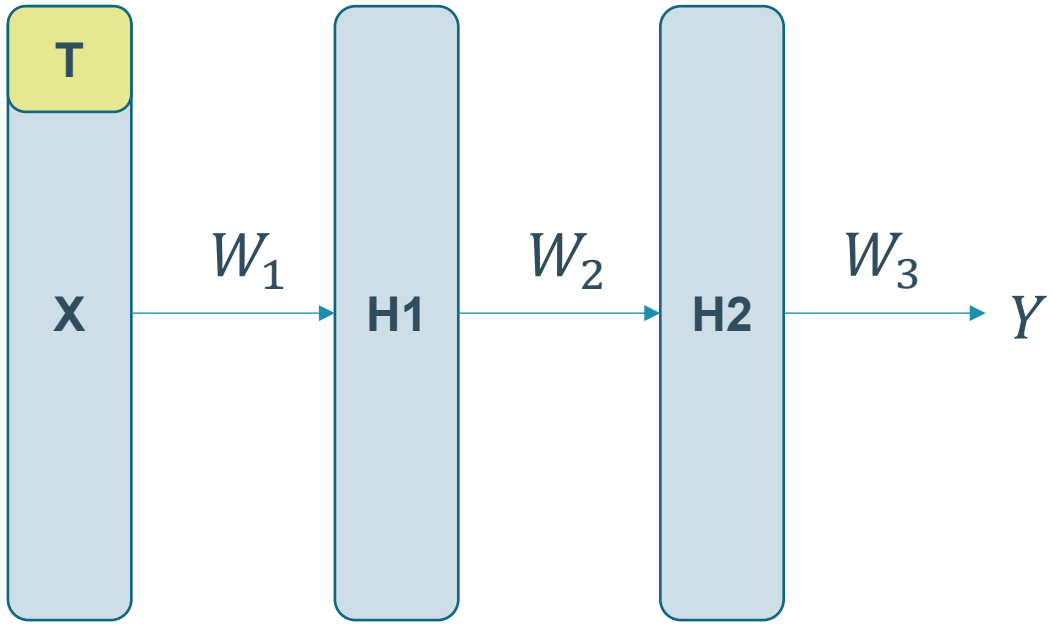
- Estimate the **transformed outcome** y' using any classification method
  - Logistic regression, XGB, ANN, …
- Accurate?

Exclude or include t as a predictor?

Figure adopted from Kane, K., Lo, V. S., & Zheng, J. (2014). Mining for the truly responsive customers and prospects using true-lift modeling: Comparison of new and existing methods. Journal of Marketing Analytics, 2, 218-238.

4.2 Artificial Neural Networks

Artificial Neural Networks



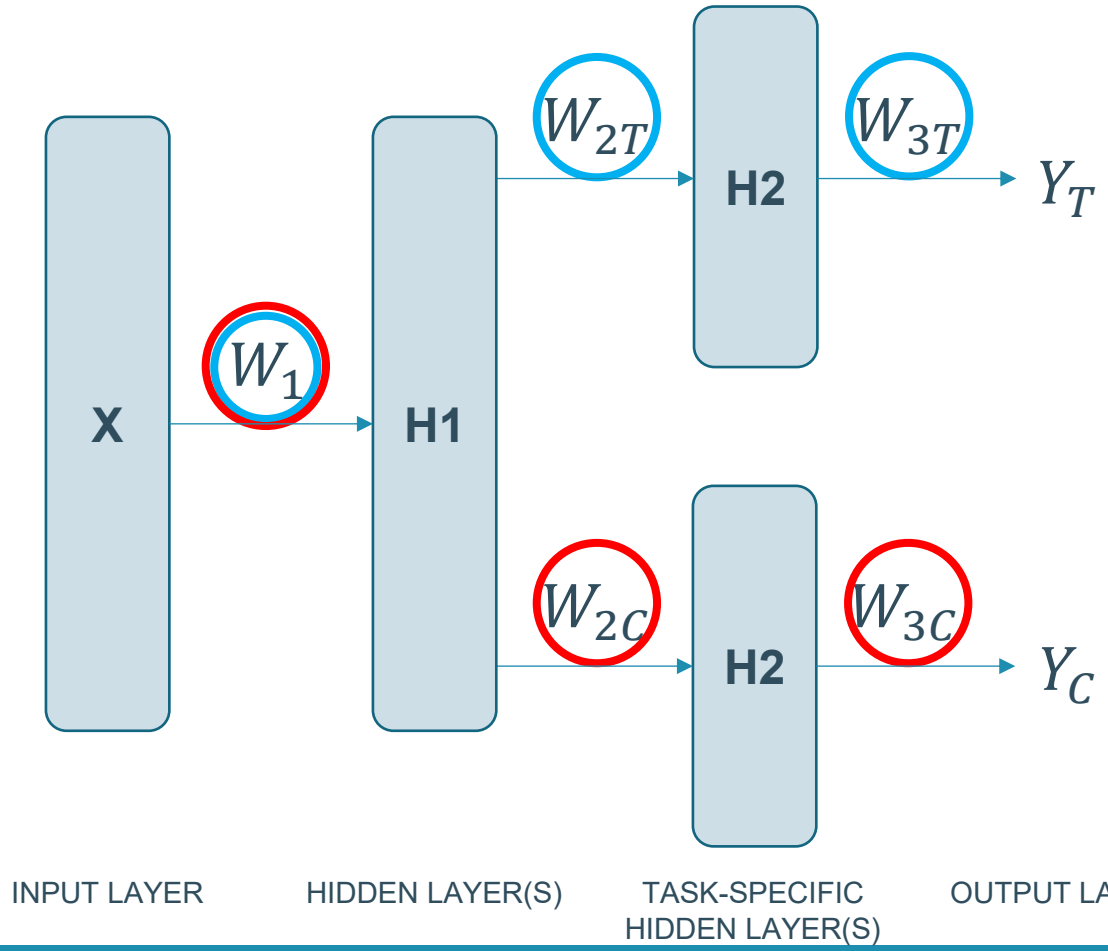
- MLP as base learner in **meta-learner**
  - E.g., S-Learner (treatment dummy)
  - $ITE = Y(T = 1) - Y(T = 0)$

"A particularly challenging aspect of training neural networks for counterfactual inference is that the ***influence of the treatment indicator variable t may be lost in high-dimensional hidden representations****"* (Schwab et al., 2020)

Schwab, P., Linhardt, L., Bauer, S., Buhmann, J. M., & Karlen, W. (2020, April). Learning counterfactual representations for estimating individual dose-response curves. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 34, No. 04, pp. 5612-5619).

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Artificial Neural Networks



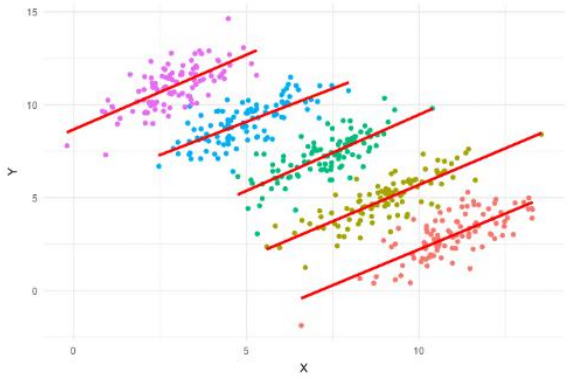
- **TARNET architecture** uses a **shared base network** and **separate head networks** for both treatment options
  - MLP-specific approach: "**Y-net**"
  - Multi-task learning
  - Hybrid **two-model architecture**
  - For binary treatment or multi-treatment

- Observations of the treatment group for learning (partial updates) W_1 and W_{2T} and W_{3T}
- Observations of the control group for learning (partial updates) W_1 and W_{2C} and W_{3C}

Shalit, U., Johansson, F. D., & Sontag, D. (2017, July). Estimating individual treatment effect: generalization bounds and algorithms. In: International Conference on Machine Learning (pp. 3076-3085). PMLR.

4.3 ANN with balancing

Learning from observational data

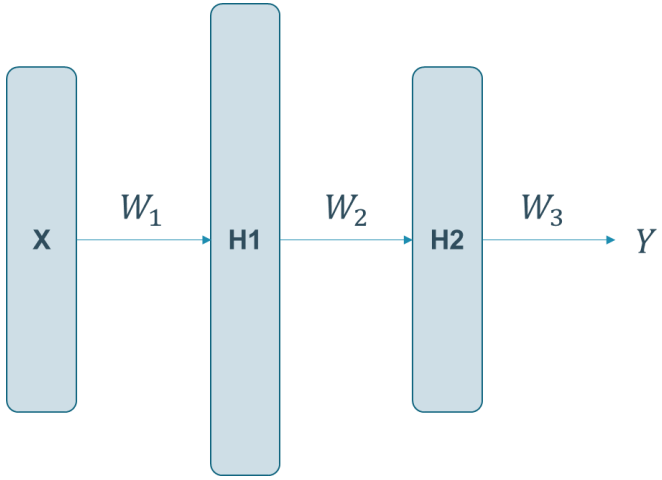


- Obtaining RCT data may be costly, unethical, time-consuming, impossible
- Learning from observational data may be a solution, but then it is important to address the potential **assignment bias** that results from using observational data
  - Imagine a clinical trial for an advanced stage cancer.
  - When contemporary medicine (say drug A) is not helping, the acting clinician may forward the patient to a running clinical trial (for drug B).
  - After the trial, we should compare outcomes, i.e., E[Y|B] vs. E[Y|A].
  - Because we did not randomly assign the new (and old) drug to patients, the expectation (E) is not calculated over the same distribution!
  - Typically, patients that were forwarded to drug B, were already in a worse state.
  - As such, we may **severely underestimate the efficacy of drug B**.
- A variety of methods have been proposed to address this selection bias
  - Weighting approaches, matching, synthetic control, …
- Most of these try to '**balance**' the control and treatment sample
  - I.e., to **make their distributions similar**

Cf. survey data analysis

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Deep learning with balancing

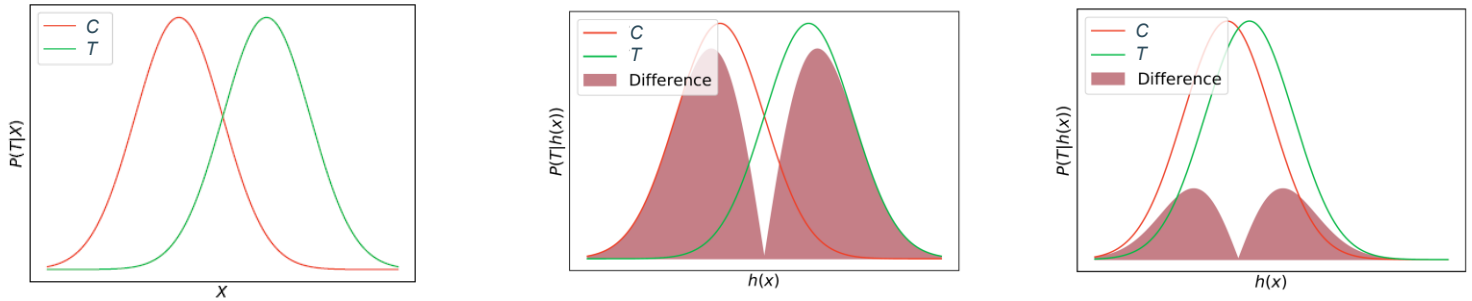


Assignment bias leads to differently distributed C and T

Hidden layer H1:

- To be interpreted as a **representation h(X) of X**
- **Idea!** Learn *bias-free* representation by minimizing **distributional distance**
- **How**?

  Step 1: Measure amount of ***bias***: Maximum Mean Discrepancy (MMD)

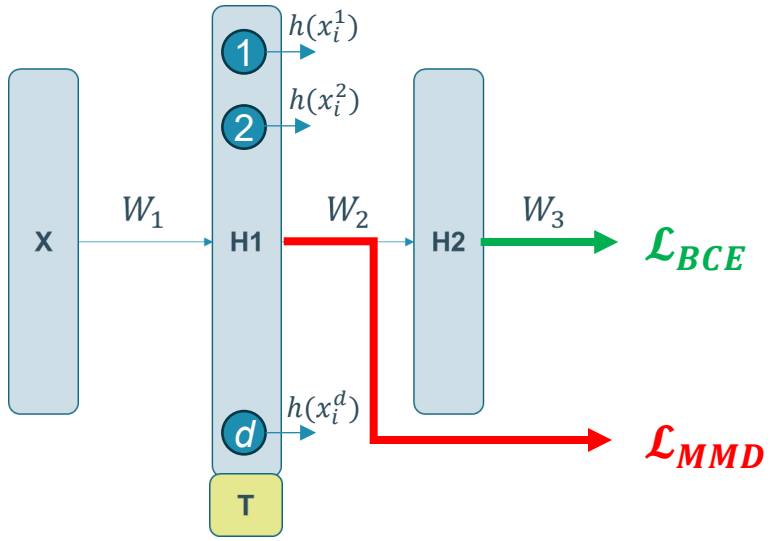


(a) Large MMD (b) Small MMD

Shalit, U., Johansson, F. D., & Sontag, D. (2017, July). Estimating individual treatment effect: generalization bounds and algorithms. In *International Conference on Machine Learning* (pp. 3076-3085). PMLR.

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Deep learning with balancing



- **Idea!** Learn bias-free representation by minimizing **distributional distance**
- **How**?

  Step 1: Measure amount of bias: Maximum Mean Discrepancy (MMD)

  Step 2: Extend loss function to minimize bias in H1:

$$Loss = \mathcal{L}_{BCE} + \alpha \cdot \mathcal{L}_{MMD}$$

Regularization scheme to address assignment bias

α : **Hyperparameter**

$\mathcal{L}_{BCE}$: **Binary cross-entropy loss (log-likelihood)**

$\mathcal{L}_{MMD}$: **Maximum mean discrepancy**

$$= \sum_{i=1}^{d} \left| \frac{1}{N_T} \sum_{i_T=1}^{N_T} h\left(x_{i_T}^{i}(t=1)\right) - \frac{1}{N_C} \sum_{i=1}^{N_C} h\left(x_{i_C}^{i}(t=0)\right) \right|$$

Sum across *d* neurons in hidden layer h

Average output value of treated

Average output value of non-treated

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Thoughts on methods

- The essence of causal machine learning for treatment effect modeling is in handling assignment or confounding bias, rather than in handling the ‘fundamental problem of causal inference’

- The ‘fundamental problem of causal inference’ is rather a problem of evaluation

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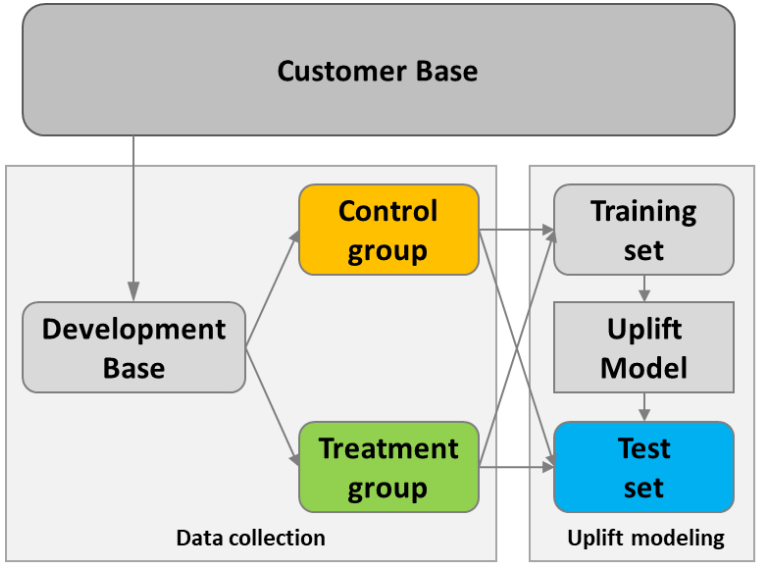
5. Evaluation

Challenges: summary

- Pre-processing:
  - Identifying hidden confounders
  - Data requirements for learning
- Processing:
  - Non-observed target variable
  - Learning from observational data
- Post-processing:
  - **Non-observed target variable**
  - Data requirements for monitoring

The fundamental problem of causal inference:

- Target variable not observed!
- No comparison of prediction vs. observed outcome!



Evaluation

- Use any evaluation measure for classification with the estimates for the assigned treatment
  - Requires RCT data! (alternative approaches, see Schuler et al.)
  - **Indirect evaluation** of CATE estimate!

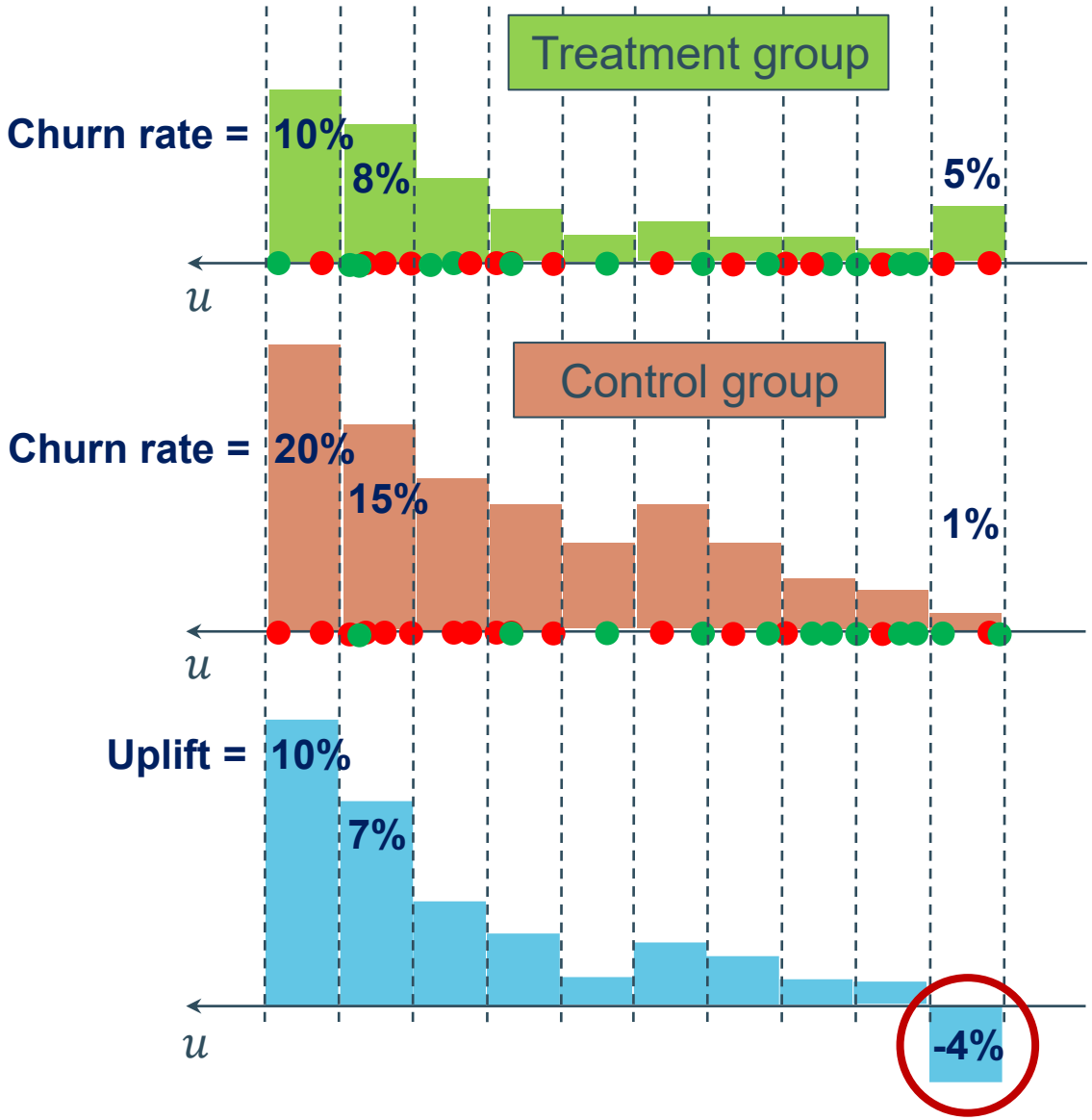
- **PEHE**: **precision in estimation of heterogeneous effect**
  - = RMSE (~root mean squared error)
  - Synthetic or semi-synthetic data (simulated data, e.g., in research; see also Ch.11, B&RL)

$$\epsilon_{PEHE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}([\underbrace{Y_1^{(i)} - Y_0^{(i)}}_{\text{ITE}}] - [\underbrace{\hat{Y}_1^{(i)} - \hat{Y}_0^{(i)}}_{\text{ITE estimate}}])^2}$$

Schuler, A., Baiocchi, M., Tibshirani, R., & Shah, N. (2018). A comparison of methods for model selection when estimating individual treatment effects. *arXiv preprint arXiv:1804.05146*.

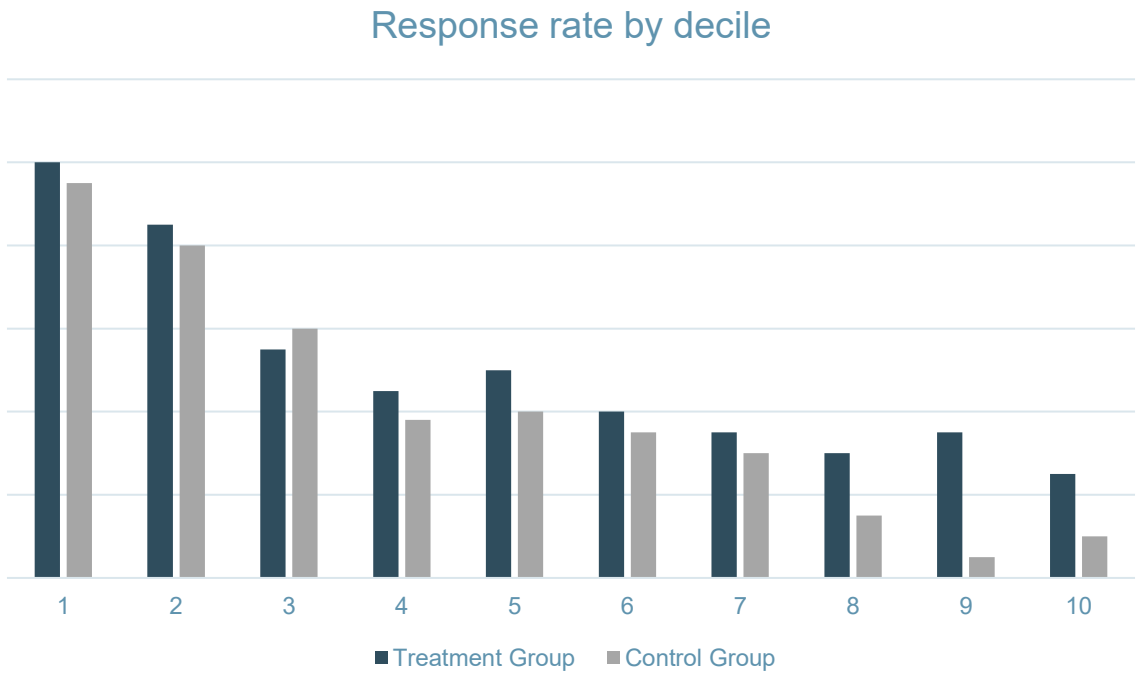
Uplift curve

- For binary target
- Evaluation by comparing outcomes for ***similar*** groups
- Uplift model allows to score and rank all instances
- **Uplift-curve:** increase in positive outcome rate
  - E.g., per decile
- Note: observational data …



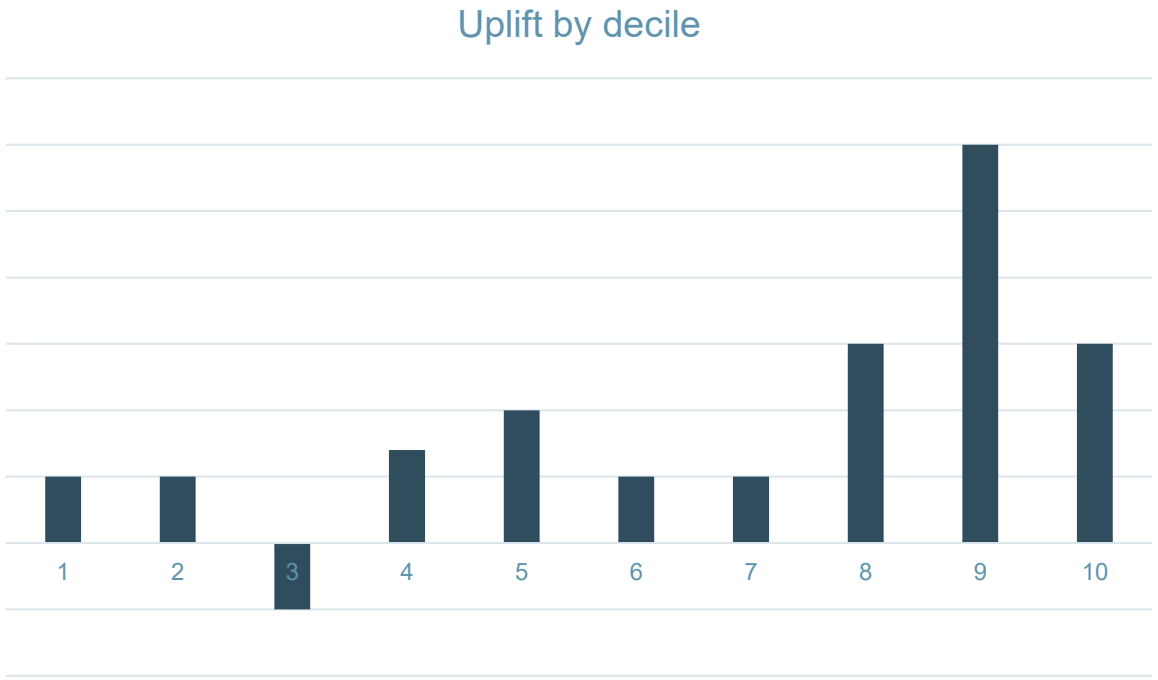
Visual evaluation approaches

- Comparison at group level



Visual evaluation approaches

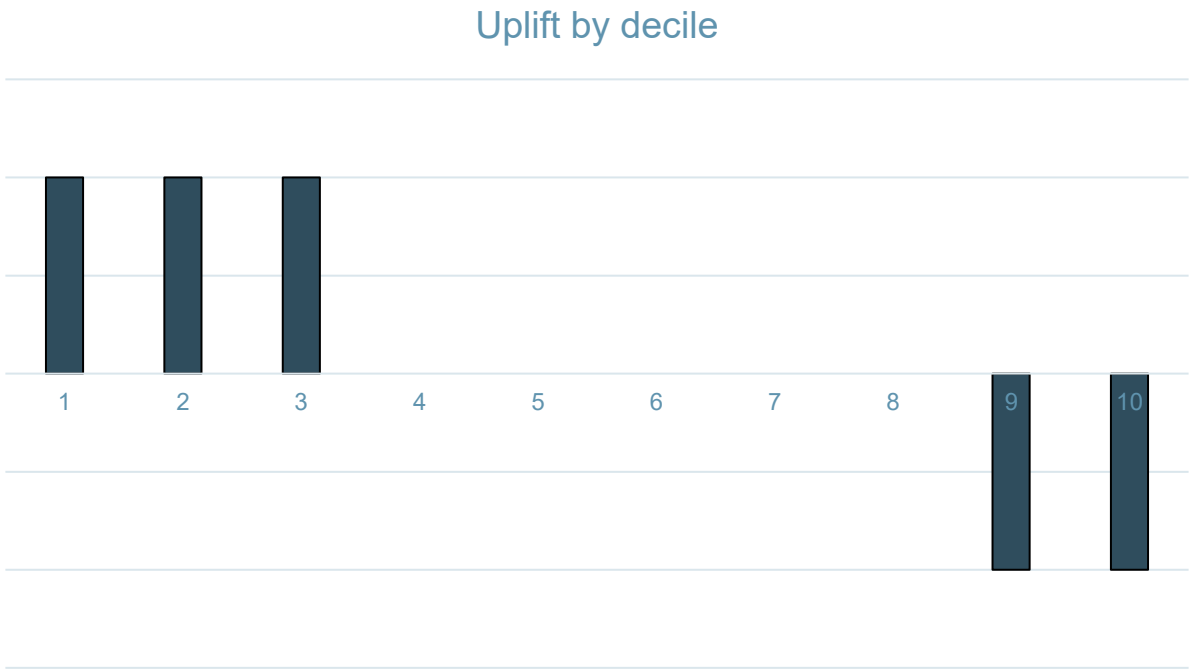
- Uplift by decile curve



Good model?

Visual evaluation approaches

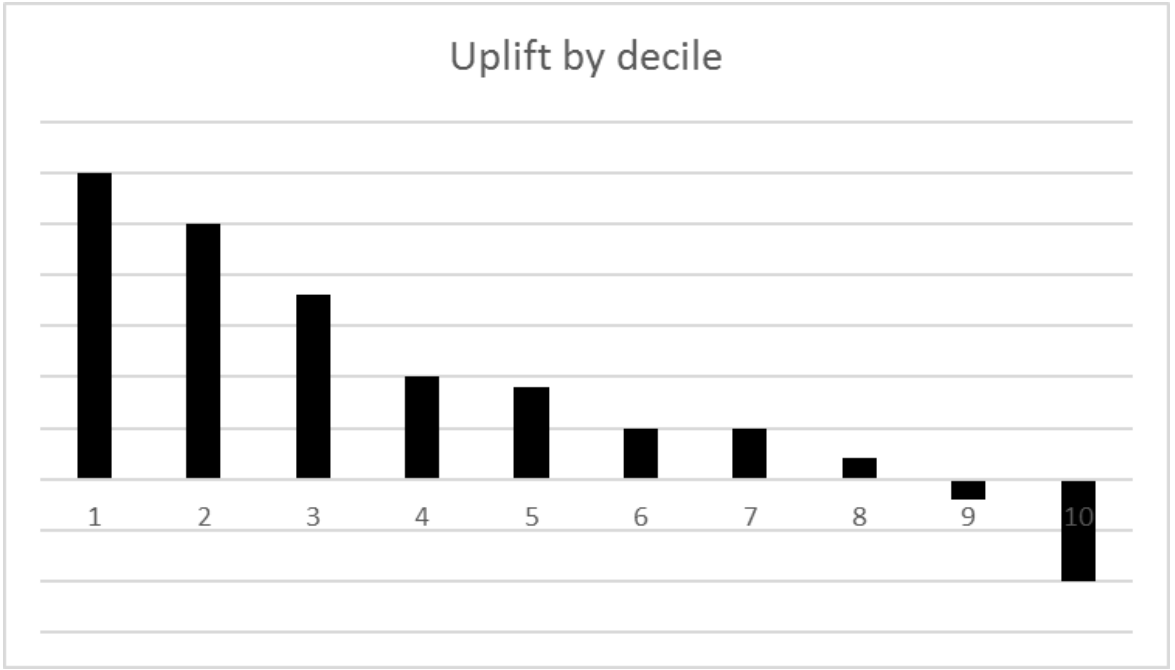
- Uplift by decile curve



Good model?

Visual evaluation approaches

- Uplift by decile curve



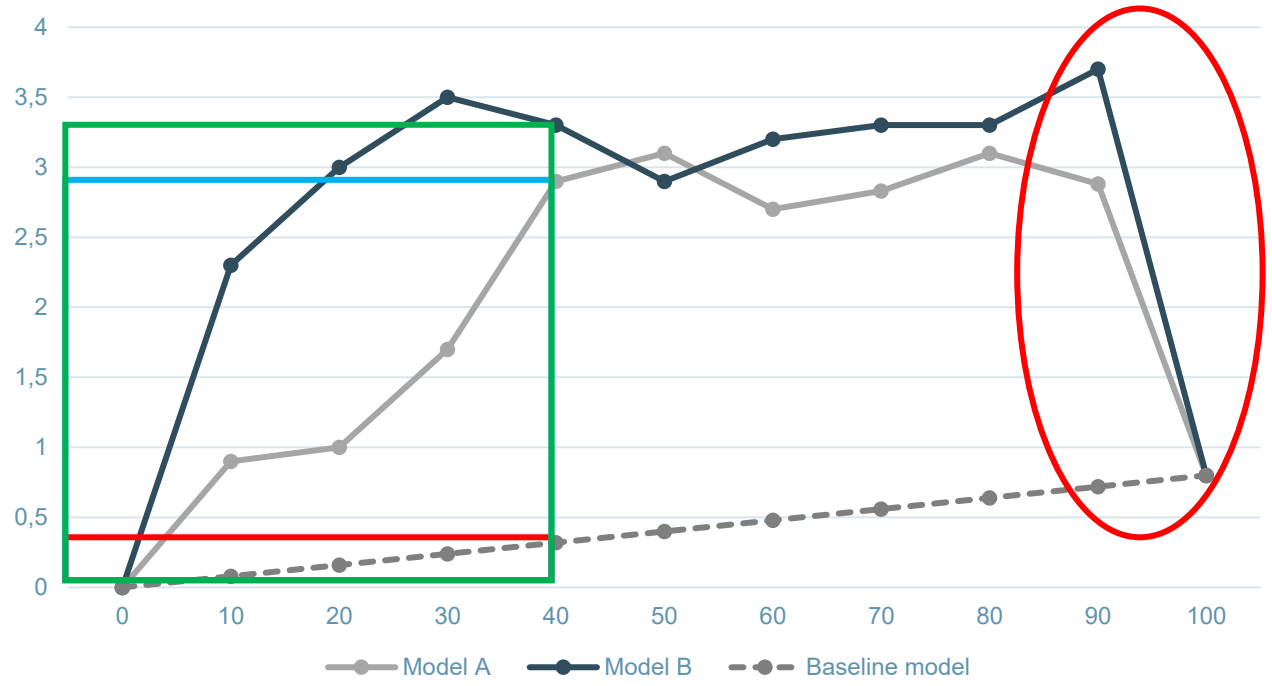
Good model?

Visual evaluation approaches

Link with ATE?

Cumulative incremental gains or Qini curve (cf. Gini curve)

Y: **Increase in positive outcome rate (%)**

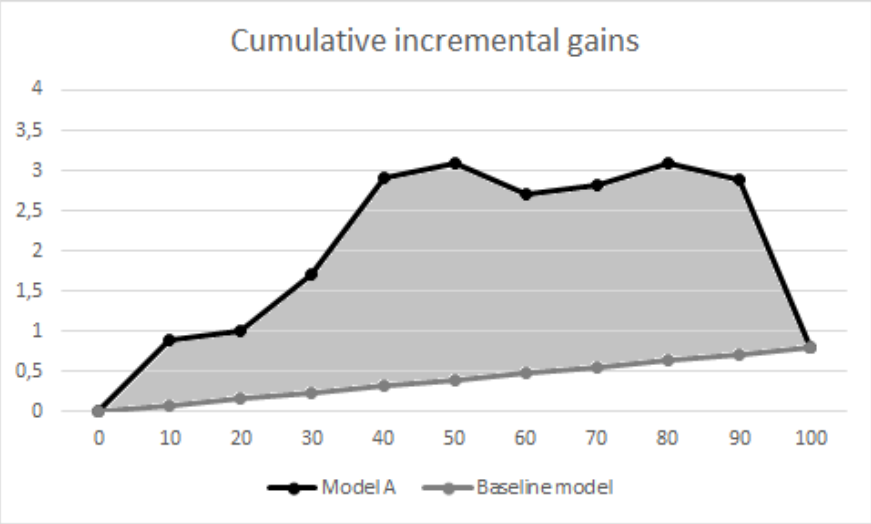


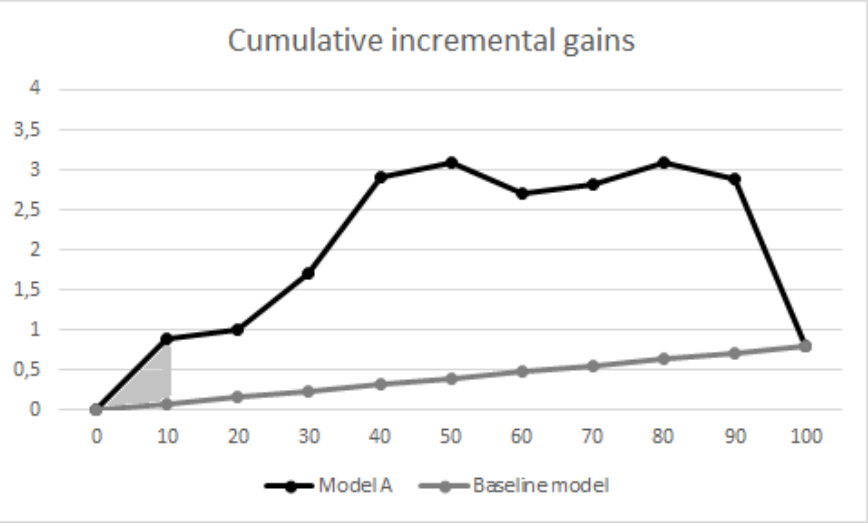
X: **Treatment rate (%)** of test sample ranked with model from large to small estimated uplift

Baseline (random) model:	40% treated	→	0.7% increase
Model A:	40% treated	→	2.9% increase
Model B:	40% treated	→	3.3% increase

Performance measures

- Qini measure = Area Under the Uplift Curve (AUUC <> AUC)
- **Quantile uplift**: how much uplift is achieved at a specified targeting depth
  - **Similar**: top-decile qini



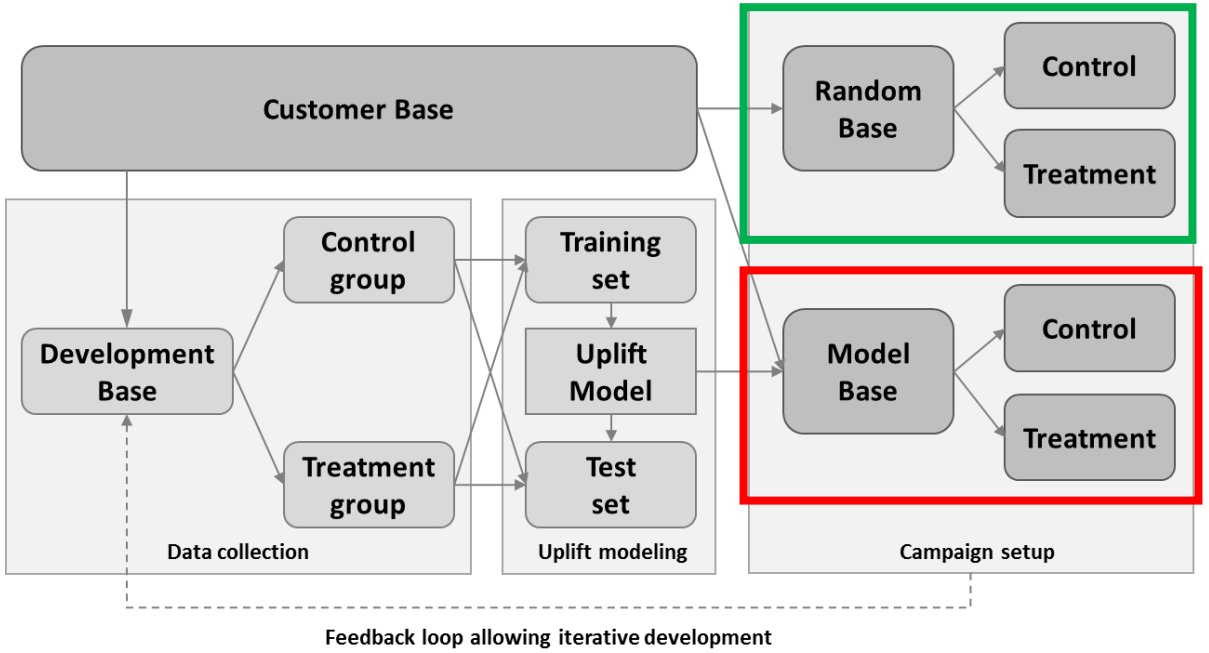


Challenges: summary

- Pre-processing:
  - Identifying hidden confounders
  - Data requirements for learning
- Processing:
  - Non-observed target variable
  - Learning from observational data
- Post-processing:
  - Non-observed target variable
  - **Data requirements for monitoring**

Challenges: post-processing

- Remember: feedback loop
- Integrate *virtuous* continuous feedback loop in model deployment!
- Iterative ***learning*** and improving or optimizing



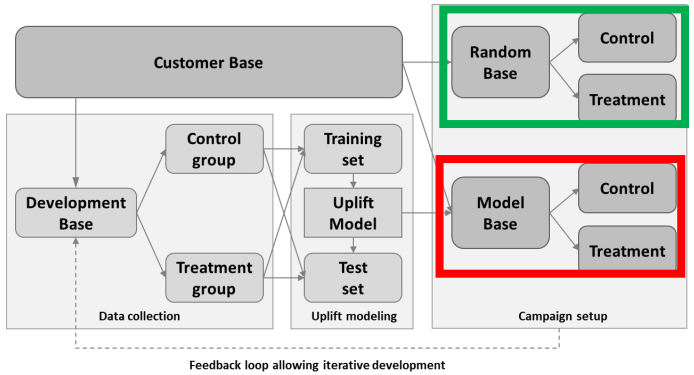
RCT

Exploration vs. Exploitation

Active learning
Reinforcement learning

Backtesting operational uplift model

- Campaign effectiveness and model performance measurement with **true lift** (Lo, 2002)



	Treatment Group	Control Group	Treatment - Control
Model Base	5.3%	1.2%	4.1%
Random Base	2.1%	0.8%	1.3%
			2.8%

= ATE

$$\text{True Lift} = (5.3\% - 1.2\%) - (2.1\% - 0.8\%) = 2.8\%$$

Effect of using the model

Effect of the model + campaign

Baseline effect of the campaign = ATE

What if we find the baseline effect of the campaign to be close to zero?

6. Cases

6.1 Case: customer churn

6.2 Case: machine maintenance

6.3 Case: acceptance risk

6.1 Case: customer churn

Case: customer churn

Analytics to the rescue?

- Data-driven solution: **customer churn prediction models**
  - Identification of churners and their motivations allows to:
    - **Target** future churners in retention campaigns
    - Increase **efficiency** of such campaigns
  - Small improvement in customer retention may yield significant profit increase
  - Ascarza et al. (2018):
    - Comprehensive framework of questions for designing effective retention strategies
    - "Who is at risk and why?"
    - "Whom should we target and with what incentives?"
    - "When do we target?"
    - "What can we gain?"

Predictive analytics (ch.2-7)

Prescriptive analytics (ch.9)

Cost-sensitive analytics (ch.11)

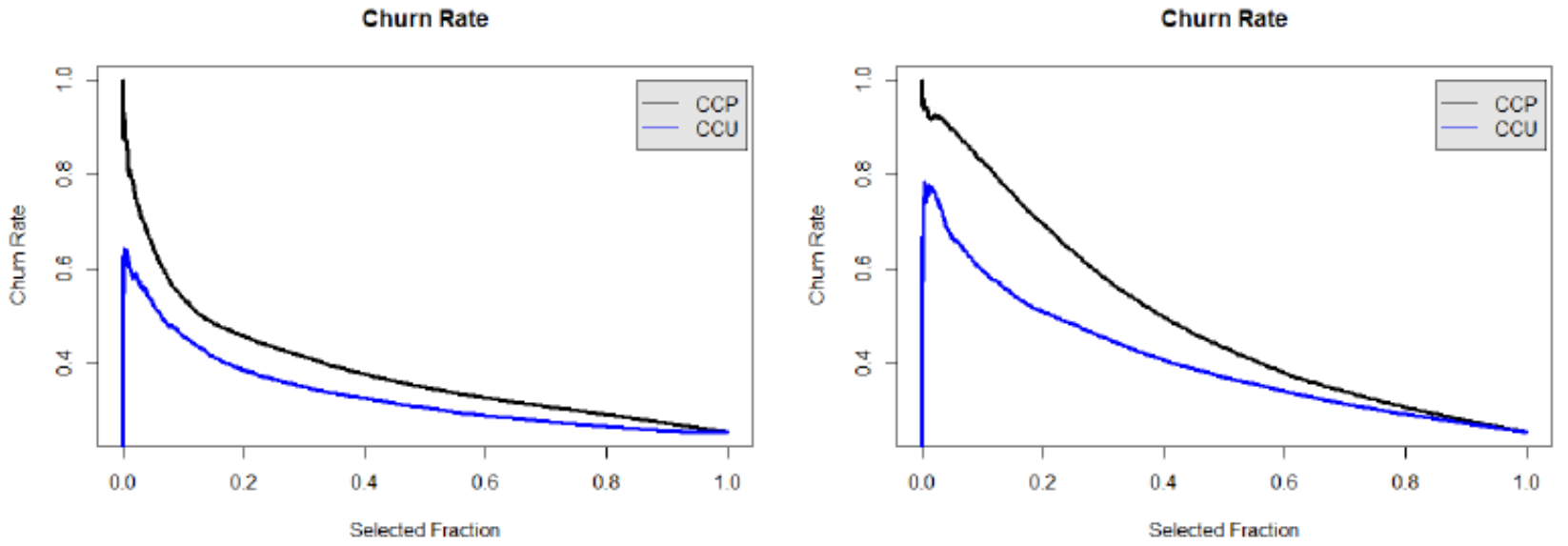
Ascarza, E. et al. (2018). In pursuit of enhanced customer retention management: Review, key issues, and future directions. *Customer Needs and Solutions*, *5*(1), 65-81.

Case: customer churn

- The proof of the pudding…
- Experiment:
  - Dataset:
    - Customer churn
    - Credit card
    - > 100,000 observations
    - Treatment and control group
  - Models:
    - Predictive models:
      - Logistic regression
      - Random forests
    - Uplift models:
      - Logistic regression with treatment dummy (Lo's) approach
      - Uplift random forests
  - Evaluation:
    - Predictive power (AUC, lift, …), Uplift (cumulative uplift), Profitability (MP measure)
    - Similarity of rankings: correlation

Case: customer churn

WINNER:
Customer churn prediction model!



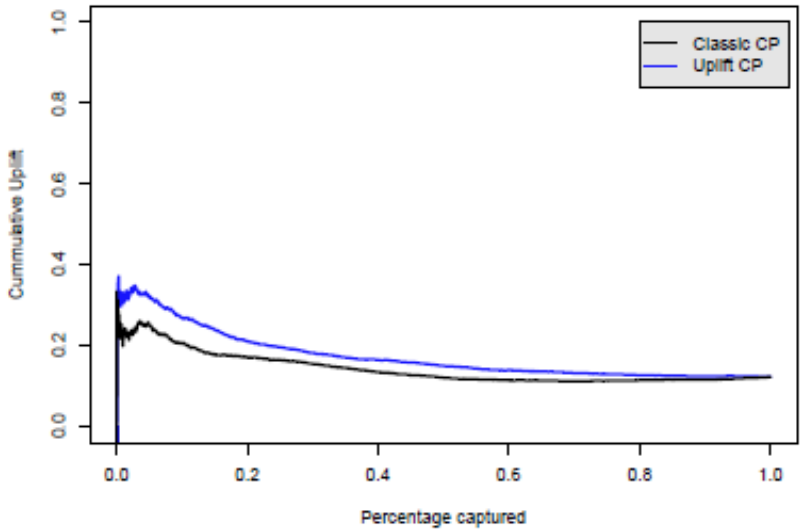
(a) Logistic Regression (b) Random Forests

Figure 8: Churn rate in function of selected fraction of customers for CCP and CCU models for both logistic regression (a) and random forests (b).

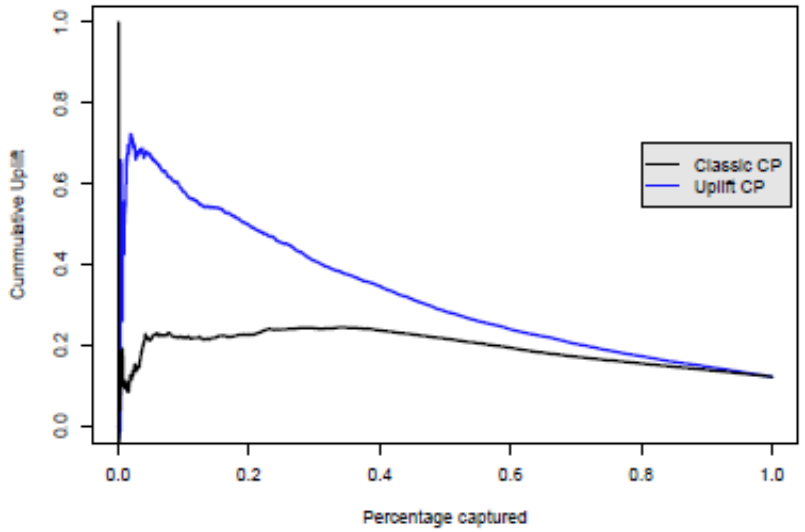
BUT! Evaluation in terms of predictive power!

Case: customer churn

WINNER:
Customer churn uplift model!



(a) Cumulative Uplift



(b) Cumulative Uplift

Figure 9: Cumulative Uplift in function of selected fraction of customers for CCP and CCU models for both logistic regression (a) and random forests (b).

Cumulative uplift = churn_rate_control(perc.capt.) – churn_rate_treatment(perc.capt.)

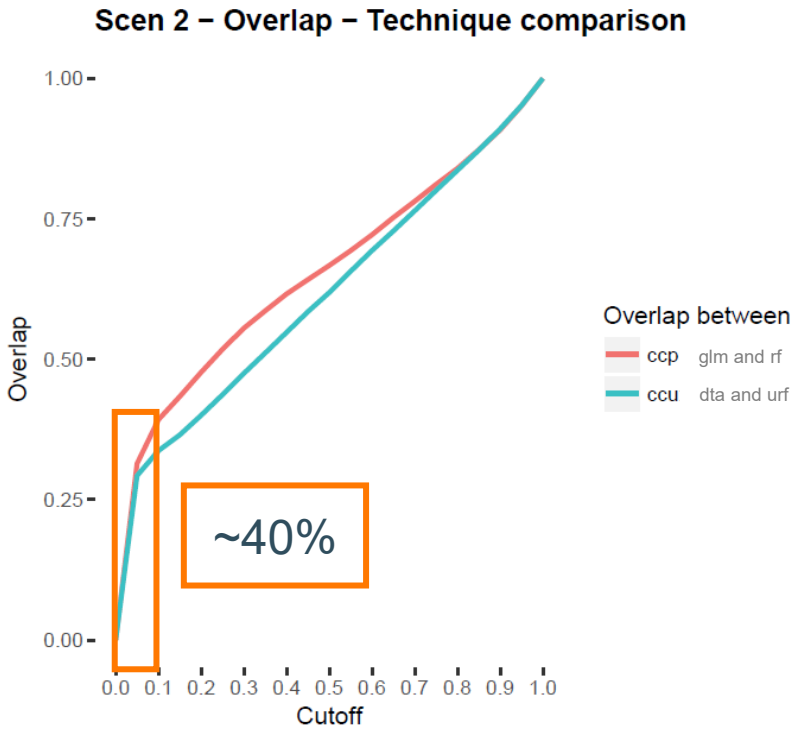
Case: customer churn

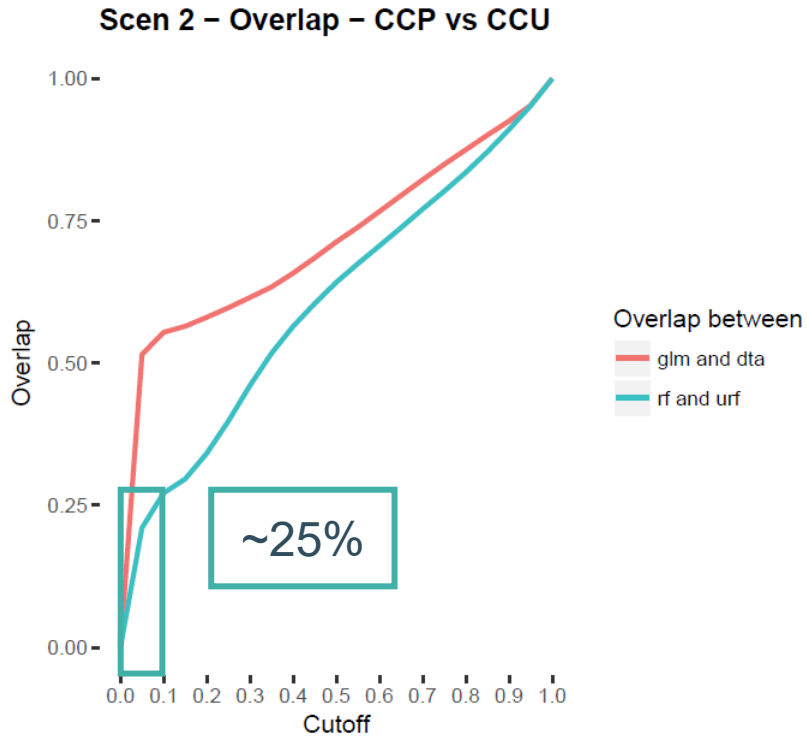
- Uplift models ***identify persuadables***, they ***do not predict churners***
  - Therefore, the churn rate is lower for the upper deciles
  - The uplift (= the reduction in churn rate), however, is significantly higher!

- Warning: **aggregated evaluation measures** may be misleading
  - 20% persuadables – 20% sleeping dogs = 0% Average Treatment Effect!
  - Torture the results, rather than the data
    - Look what's underneath aggregate performance measures!
    - How? Visually! Curves!

Case: customer churn

Comparison rankings: if you select for “**two methods**” the 10% highest scored: what part of those are the same?





2 methods: Logistic and random forests prediction
2 methods: Logistic and random forests uplift

2 methods: Logistic prediction and uplift
2 methods: Random forest prediction and uplift

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Case: customer churn

- **Conclusion**: switch from predictive to prescriptive?
- Ascarza et al. (2018): comprehensive framework of questions for designing effective retention strategies
  - "Who is at risk and why?"
  - "Whom should we target and with what incentives?"
  - "When do we target?"
  - "What can we gain?"
- Complementary!
  - How to optimally combine?
  - For predictive: use ***unbiased data***!

Predictive analytics (ch.2-7)

Prescriptive analytics (ch.9)

Cost-sensitive analytics (ch.11)

6.2 Case: machine maintenance



Case: machine maintenance

Challenge: the expenses associated with maintenance (failures, overhauls, & preventive maintenance account for 15–40% of production costs (Dunn, 1987; Lofsten, 2000)

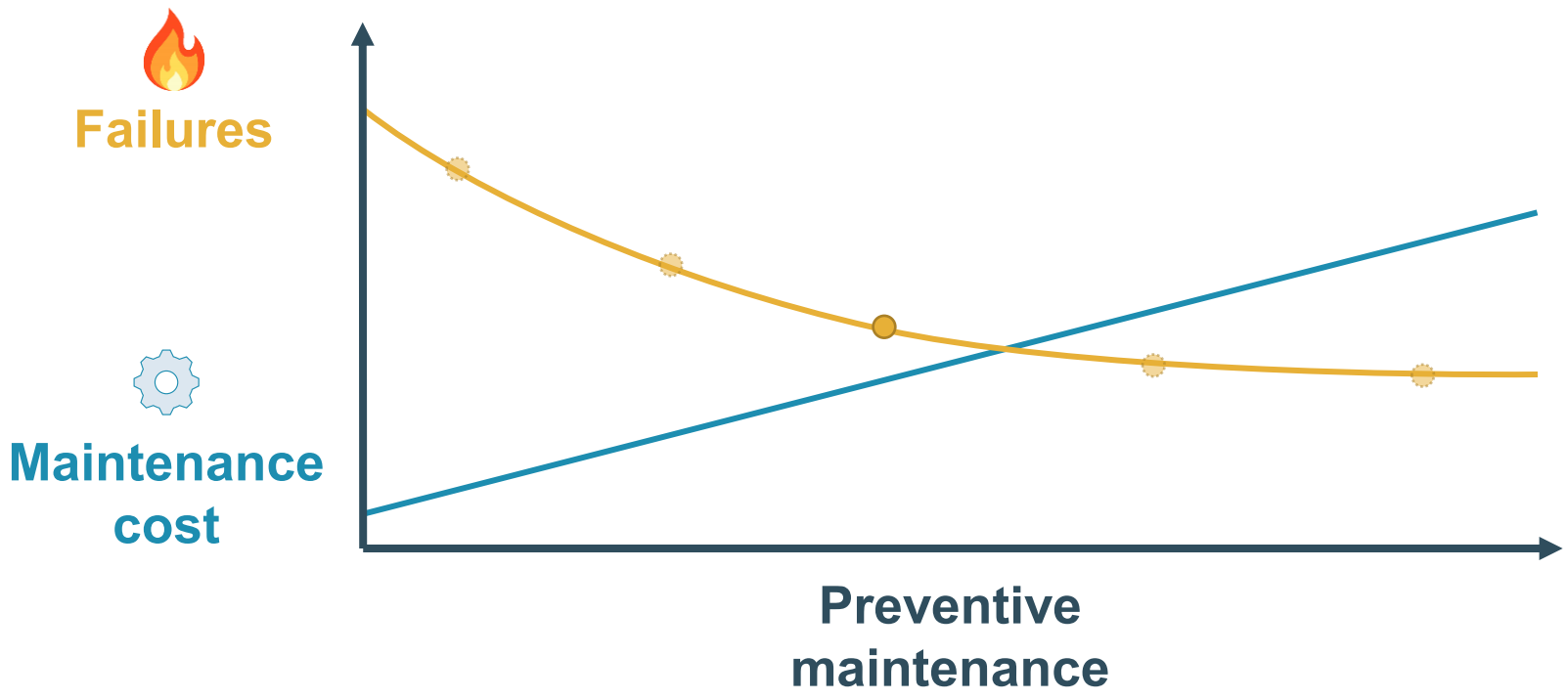
Goal: minimize costs related to:

- **Machine failures and overhauls**
- **Preventive maintenance**

- ***Overhauls*** are unplanned, comprehensive maintenance interventions during which large parts of the machinery need to be replaced. From the viewpoint of the full-service maintenance provider, these are the most costly type of event.
- ***Failures*** are also unplanned and result in an urgent need for maintenance as the machine stops running until corrective maintenance occurs. A failure also incurs a cost to the service provider that is smaller than the cost of an overhaul, but larger than the cost of PM.

R. Dunn. Advanced maintenance technologies, Plant Engineering, 40 (12) (1987), pp. 80-82
Lofsten, H. (2000) Measuring maintenance performance – in search for a maintenance productivity index, IJPE, Vol.63, pp. 47-58.

Case: machine maintenance



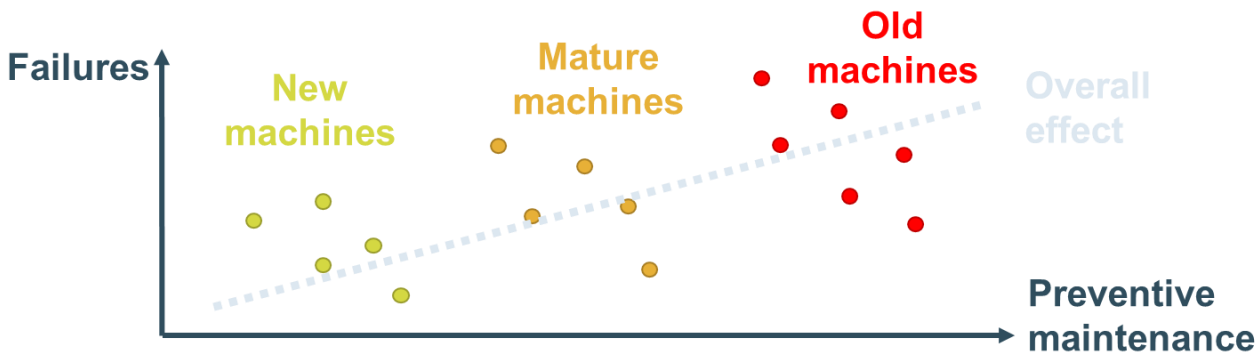
Case: machine maintenance

- **Problem:** prior to contract start, **decide on preventive maintenance frequency**
  - **Machine** $x_i \in \mathbb{R}^d$
    - x_i: machine type, age, industry, etc.
  - **Preventive maintenance frequency** $t_i \in \mathbb{R}^+$
  - **Outcomes**:
    - Overhaul rate $o_i \in \mathbb{R}^+$
    - Failure rate $f_i \in \mathbb{R}^+$
- **Objective**:
  - Given machine x_i
  - **Find optimal t_i^* to minimize costs of t_i^*, o_i and f_i**

Methodology

1. Predict overhauls $o_i(t_i)$ and failures $f_i(t_i)$ as a function of the preventive maintenance frequency t_i using **observational data**
   - SCIGAN
2. Decide on optimal PM frequency t_i^* to minimize expected total cost:

$$c(t_i) = c_t t_i + c_o o_i(t_i) + c_f f_i(t_i)$$



Bica, I., Jordon, J., & van der Schaar, M. (2020). Estimating the effects of continuous-valued interventions using generative adversarial networks. *Advances in Neural Information Processing Systems*, *33*, 16434-16445.

SCIGAN

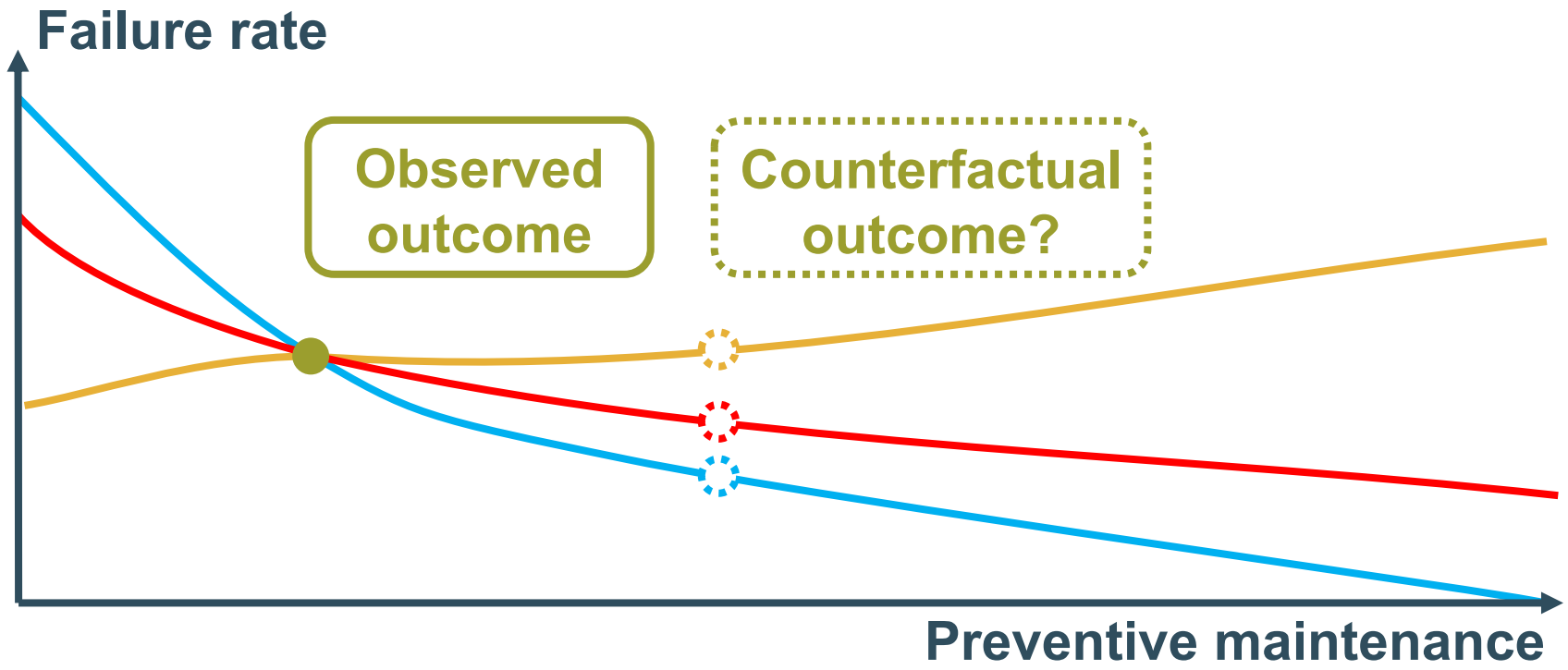
1. Predict potential outcomes $o_i(t_i)$ and $f_i(t_i)$ with **SCIGAN** (Bica et al., 2020)



Bica, I., Jordon, J., & van der Schaar, M. (2020). Estimating the effects of continuous-valued interventions using generative adversarial networks. *Advances in Neural Information Processing Systems*, *33*, 16434-16445.

SCIGAN

1. Predict potential outcomes $o_i(t_i)$ and $f_i(t_i)$ with **SCIGAN** (Bica et al., 2020)



Bica, I., Jordon, J., & van der Schaar, M. (2020). Estimating the effects of continuous-valued interventions using generative adversarial networks. *Advances in Neural Information Processing Systems*, *33*, 16434-16445.

SCIGAN

1. Predict potential outcomes $o_i(t_i)$ and $f_i(t_i)$ with **SCIGAN** (Bica et al., 2020)

"… presents a significant modification to the original ***GAN framework***

- *rather than the discriminator discriminating between entirely real or entirely fake samples,*
- *the discriminator is attempting* ***to identify the real component from***
  - ***a vector containing the real (factual) outcome from the dataset***
  - ***and the fake (counterfactual) outcomes generated by the generator.***"

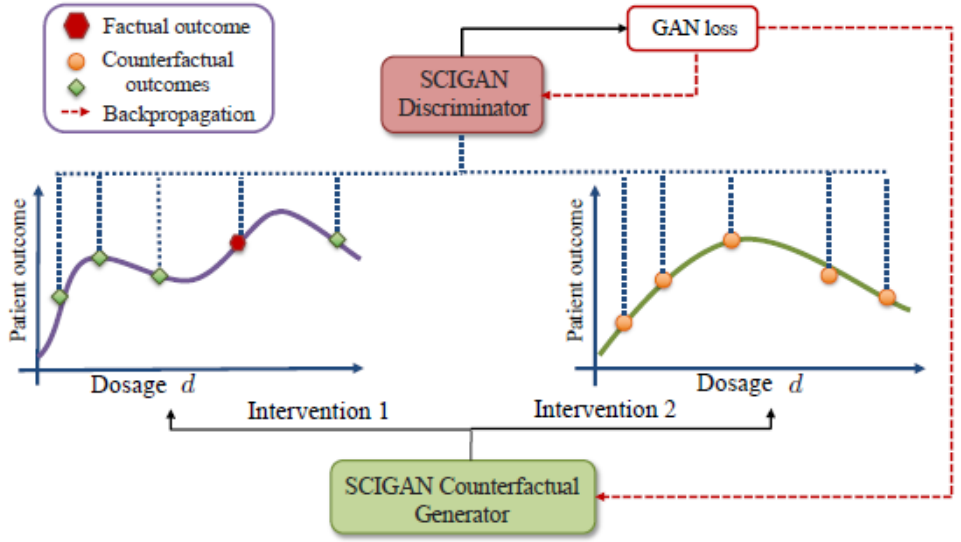


Figure 1: Overview of GAN framework used as part of SCIGAN for learning the distribution of the counterfactual outcomes.

Bica, I., Jordon, J., & van der Schaar, M. (2020). Estimating the effects of continuous-valued interventions using generative adversarial networks. *Advances in Neural Information Processing Systems*, *33*, 16434-16445.

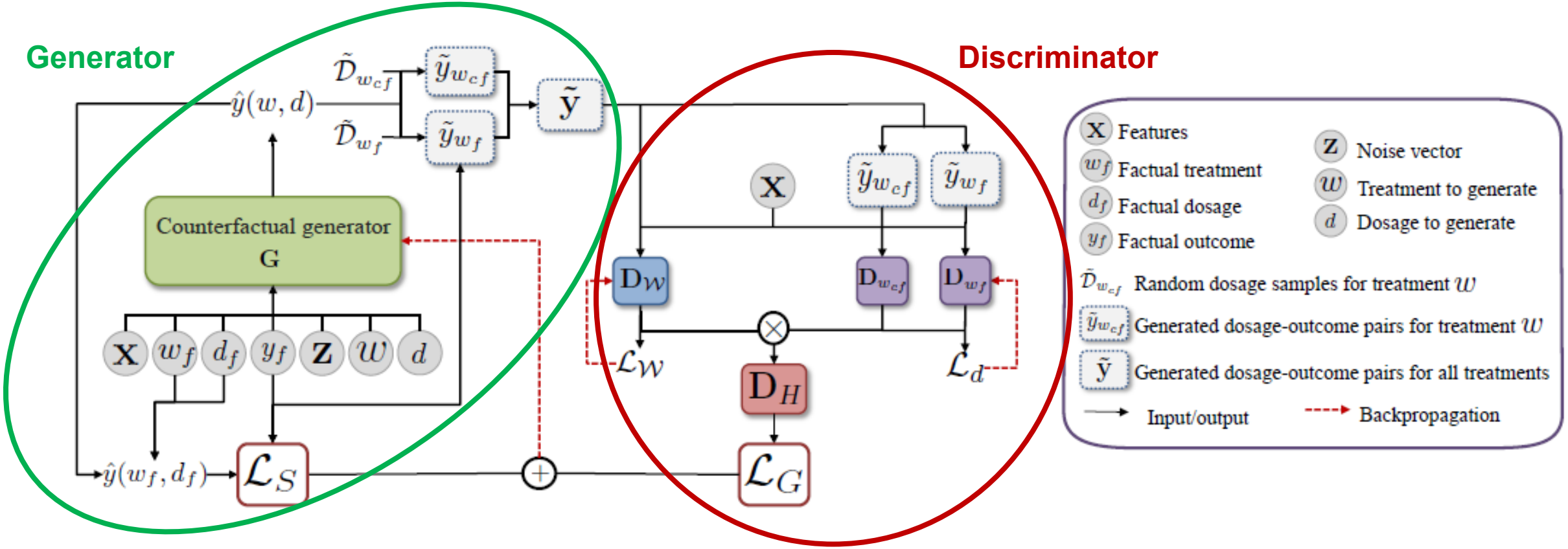


Figure 2: Overview of our model for the setting with two treatments (w_f/w_{cf} being the factual/counterfactual treatment). The generator is used to generate an output for each dosage level in each $\tilde{\mathcal{D}}_w$, these outcomes together with the factual outcome, y_f, are used to create the set of dosage-outcome pairs, $\tilde{\mathbf{y}}$, which is passed to the treatment discriminator. Each dosage discriminator receives only the part of $\tilde{\mathbf{y}}$ corresponding to that treatment, i.e. $\tilde{\mathbf{y}}_w$. These discriminators are combined (Eq. 7) to define $\mathbf{D}_H$ which gives feedback to the generator.

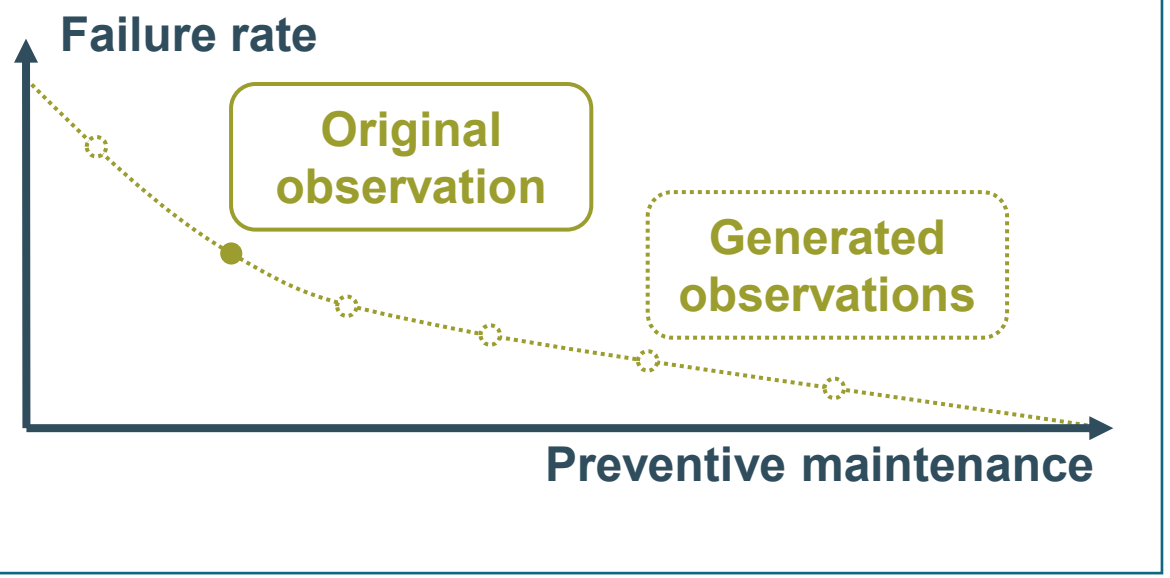
Bica, I., Jordon, J., & van der Schaar, M. (2020). Estimating the effects of continuous-valued interventions using generative adversarial networks. *Advances in Neural Information Processing Systems*, *33*, 16434-16445.

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SCIGAN

1. Predict potential outcomes $o_i(t_i)$ and $f_i(t_i)$ with **SCIGAN** (Bica et al., 2020)

1.1 Generate potential outcomes using a GAN



1.2 Supervised learning

Learn potential outcomes

- $(x_i, t_i, f_i(t_i)) \ \forall \ t_i, i$
- $(x_i, t_i, o_i(t_i)) \ \forall \ t_i, i$

Bica, I., Jordon, J., & van der Schaar, M. (2020). Estimating the effects of continuous-valued interventions using generative adversarial networks. *Advances in Neural Information Processing Systems*, *33*, 16434-16445.

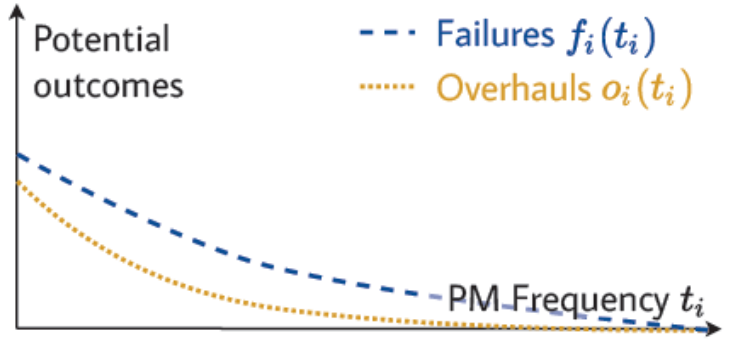
Prescriptive machine maintenance

2. Decide on optimal PM frequency t_i^* to minimize expected total cost:

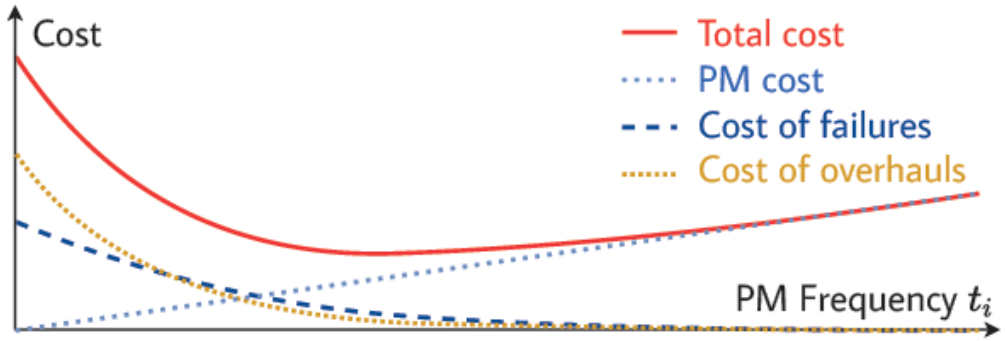
$$c(t_i) = c_t t_i + c_o o_i(t_i) + c_f f_i(t_i)$$

Feature	Value
Machine type	2
Age	10.2
Running hours	2,000
Contract type	1
...	

1. Machine information $\mathbf{x}_i$



2. Predict potential outcomes $o_i(t_i)$ and $f_i(t_i)$

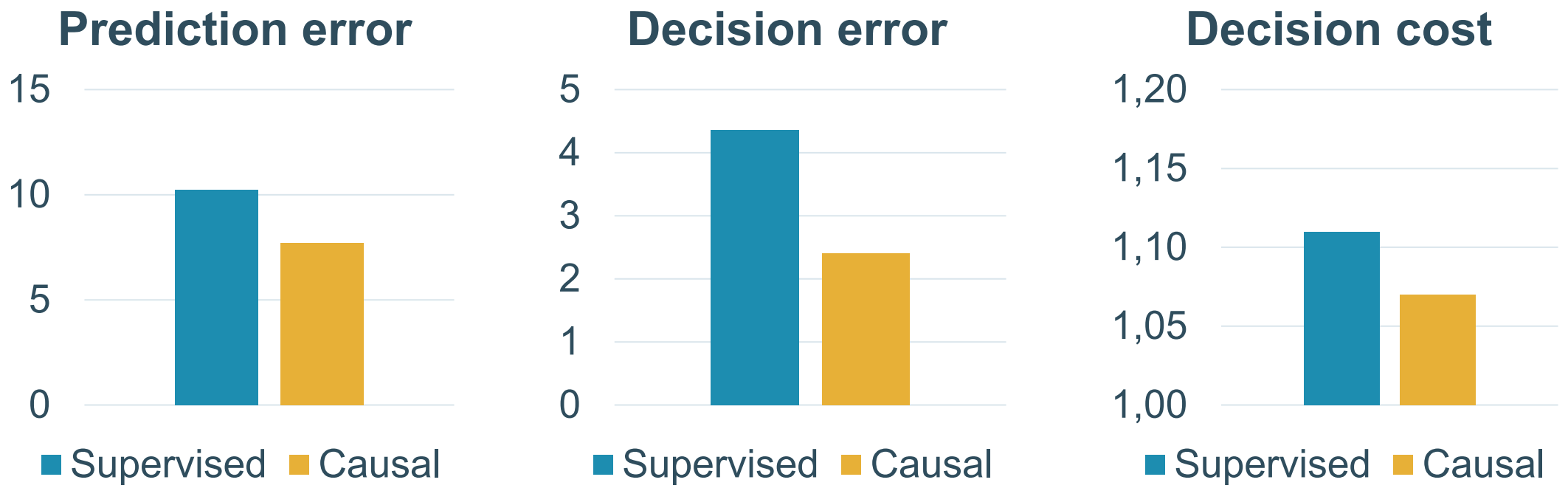


3. Prescribe the PM frequency t_i to minimize the total cost

Vanderschueren, T., Boute, R., Verdonck, T., Baesens, B., & Verbeke, W. (2023). Optimizing the preventive maintenance frequency with causal machine learning. *International Journal of Production Economics*, *258*, 108798.

Prescriptive machine maintenance

Causal approach leads to more accurate predictions and better decisions



6.3 Case: acceptance risk

Case: acceptance risk

RCT data?

- Current approach for loan pricing: pricing **grid** – i.e., segmentation based on
  - Demand characteristics (when, #, …)
  - Product characteristics (type of loan, duration, …)
  - Customer characteristics (?)
- **Idea: CATE model for modeling customer price elasticity?**
- **Note**:
  - Infeasible to price at the individual level?
    - CATE estimates still allow to optimize segmentation
  - Also applicable to insurance products, …?
    - Fences needed! (time, channel, …)

Case: acceptance risk

Maximize expected returns: Cost-sensitive decision-making (cf. Chapter 11)

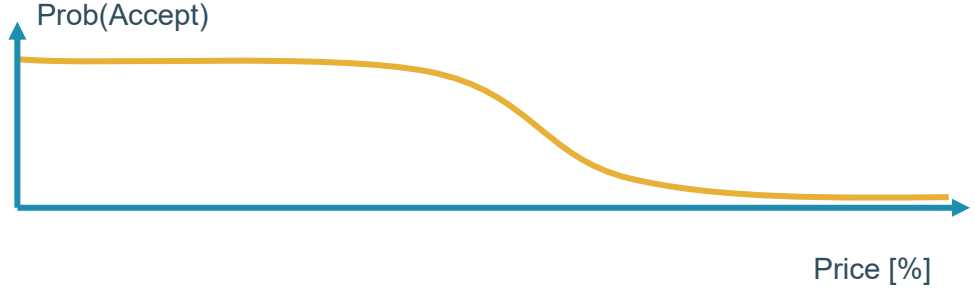
- We want an **individual price elasticity model**
  - which for any combination of **price**, **customer** and **product** characteristics,
  - a probability for the customer to accept is to be predicted
  - which allows to set the optimal price
  - so as to **maximize expected returns**
- Note: **price is a continuous treatment**
  - Methods?

Inputs:

[Age], [Job], [Income] [Duration]	Price: 3.5% p.a.	Accepted
[Age], [Job], [Income] [Duration]	Price: 5.5% p.a.	Not accepted
[Age], [Job], [Income] [Duration]	Price: 2.5% p.a.	Accepted
…	…	…
[Age], [Job], [Income] [Duration]	Price: 4.0% p.a.	Not accepted

Desired model:

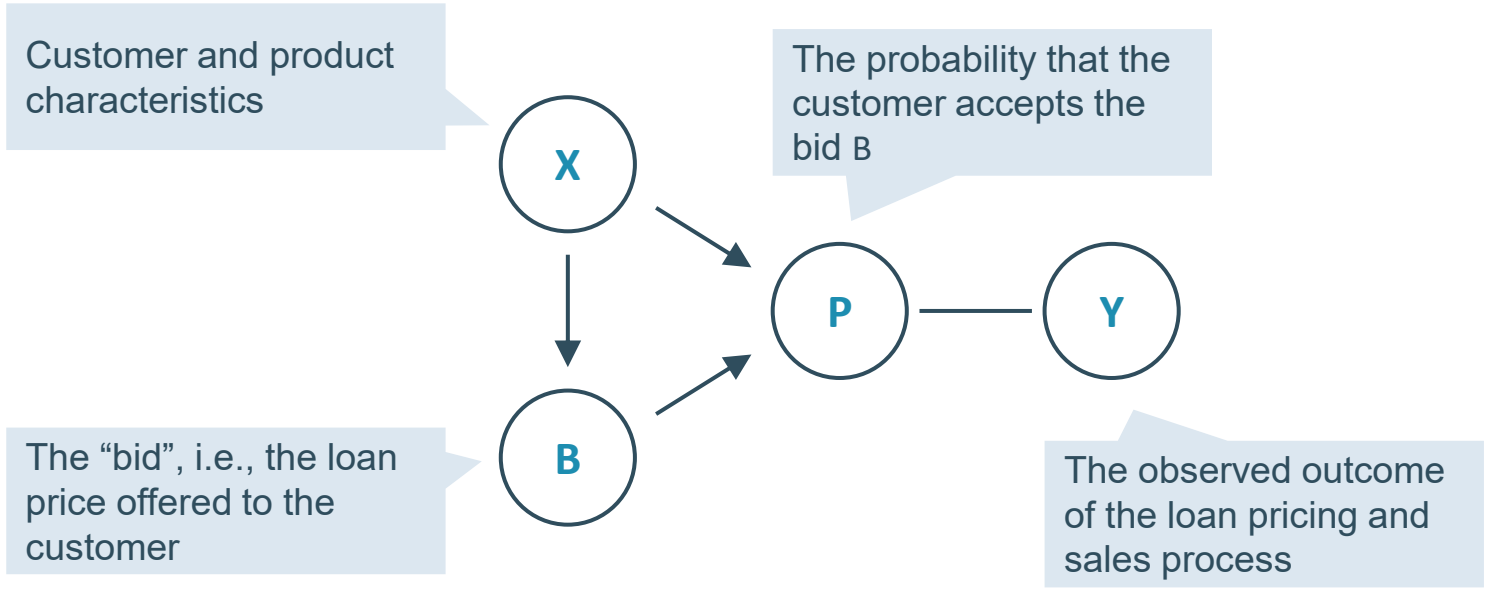
For a certain customer



Case: acceptance risk

- We want to understand the response of a customer to a loan offer (the “bid”)

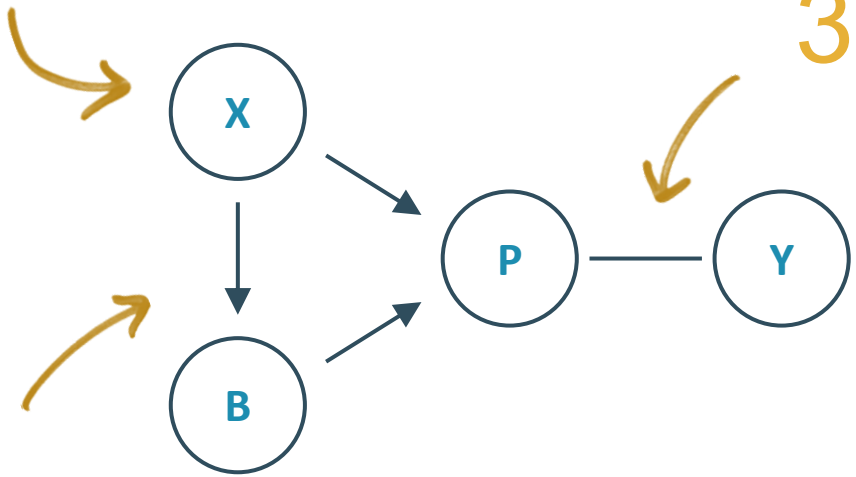
  → **Consider the following DAG**



Case: acceptance risk

- Three issues in learning a bid response from observational data:

1 Not all customer groups might be equally present in the training data



3 We are interested in **P**, but only observe **Y ~ Bernoulli(P)**

2 The bid assignment mechanism is partially unknown*

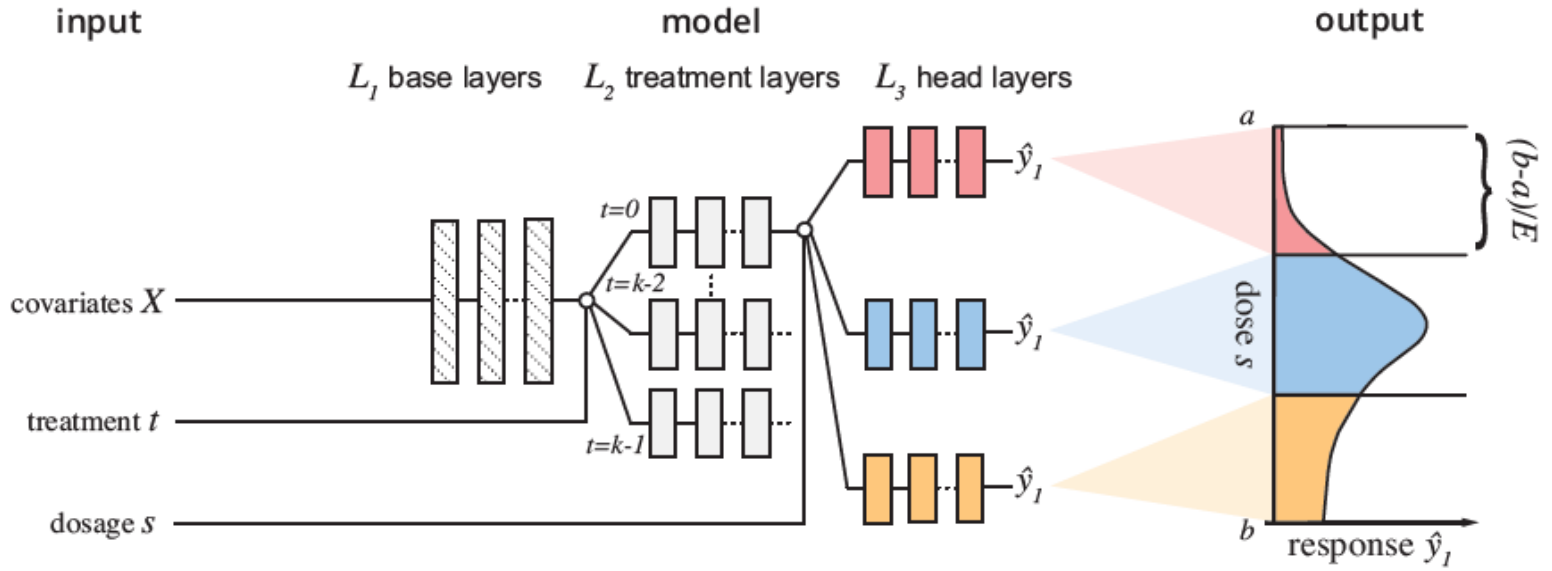
* This mechanism is driven by several factors: first, the established base price policy, e.g., a pricing grid; second, a customized price component determined by a bank clerk. This second component is driven by corporate incentive schemes, but also by the bank clerk, which is not known ex-ante and must be estimated.

Dose-Response Networks

*"A particularly challenging aspect of training neural networks for counterfactual inference is that the **influence** of the treatment indicator **variable t** may be **lost in high-dimensional hidden representations**"* (Schwab et al., 2020)

- Solved with multiple heads for categorical treatment (cf. Section 4: TARNET architecture)
- How to resolve with a **continuous treatment?**
  - **DR-net** (dose-response networks)

- Combined with **regularisation schemes (e.g., balancing)** to address **assignment bias**
  - Distribution matching (Shalit, et al. 2017)
  - Propensity dropout (Alaa, et al. 2017)



Schwab, P., Linhardt, L., Bauer, S., Buhmann, J. M., & Karlen, W. (2020, April). Learning counterfactual representations for estimating individual dose-response curves. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 34, No. 04, pp. 5612-5619).

Results show adverse effects of selection bias with beneficial performance of causal methods

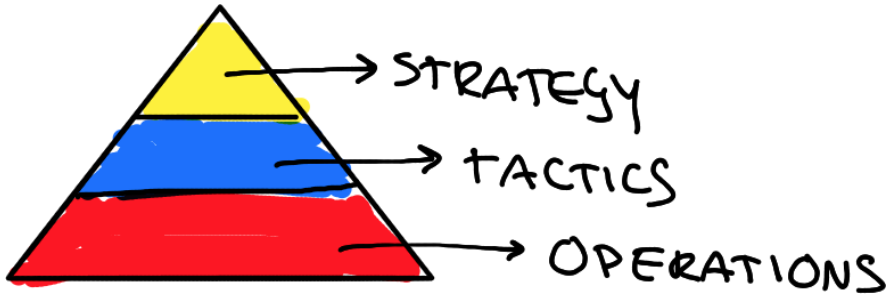
MISE over different levels of selection bias

	Bias strength				
Method	**0.0**	**5.0**	**10.0**	**20.0**	**OG[1]**
Log. Regression	0.110	0.134	0.147	0.155	*0.119*
Random Forest	0.088	0.151	0.174	0.196	0.196
MLP	**0.059**	*0.100*	*0.119*	*0.147*	0.132
GPS	0.087	0.109	0.121	0.157	0.176
DRNet	*0.075*	**0.098**	**0.115**	**0.109**	**0.103**
VCNet	0.099	0.136	0.156	0.178	0.120

- When data is unbiased, simple Neural Network (MLP) achieves best performance
- When bias increases, MLP and other non-causal methods adversely affected
- DRNet shows best performance when bias increases
- GPS and VCNet similarly affected by bias, not outperforming non-causal benchmarks

7. Concluding remarks

Prescriptive analytics



Objective of analytics: **optimizing** operational decisions

- Operations research (OR) – *Prescriptive analytics*
  - Using **mathematical modeling** for developing **simulation models** to **optimize** decision variables
  - www.orbel.be - www.euro-online.org

$Y = f(T)$

$T^* = \text{argmax}(Y)$

Argmax is an operation that finds the argument that gives the maximum value from a target function

- Predictive analytics
  - Using **data and machine learning** for developing predictive models

$Y = f(X)$

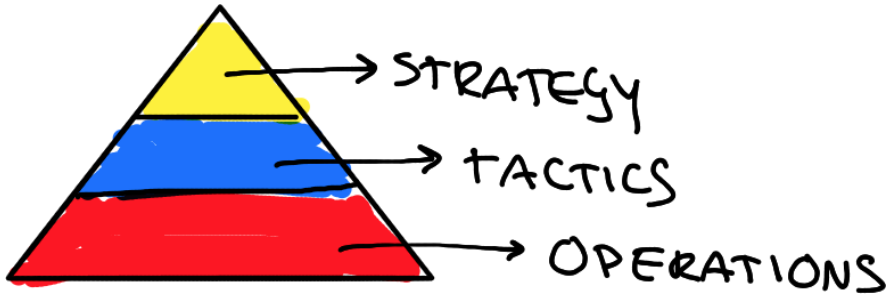
- Bridge? Prescriptive **data** analytics?
  - Can data tell us what the optimal decision is?

$Y = f(X,T)$

$T^* = \text{argmax}(Y|X)$

Prescriptive analytics

Objective of analytics: **optimizing** operational decisions



f

- Model (simulation, classification, regression, …), program, function, equation (SEM), …
  - If the result of mathematical modeling by expert: **operations research**
  - If learned by an algorithm from data: **machine learning**, **statistics**, …

$Y = f(T)$
$T^* = \text{argmax}(Y)$

Argmax is an operation that finds the argument that gives the maximum value from a target function

$Y = f(X)$

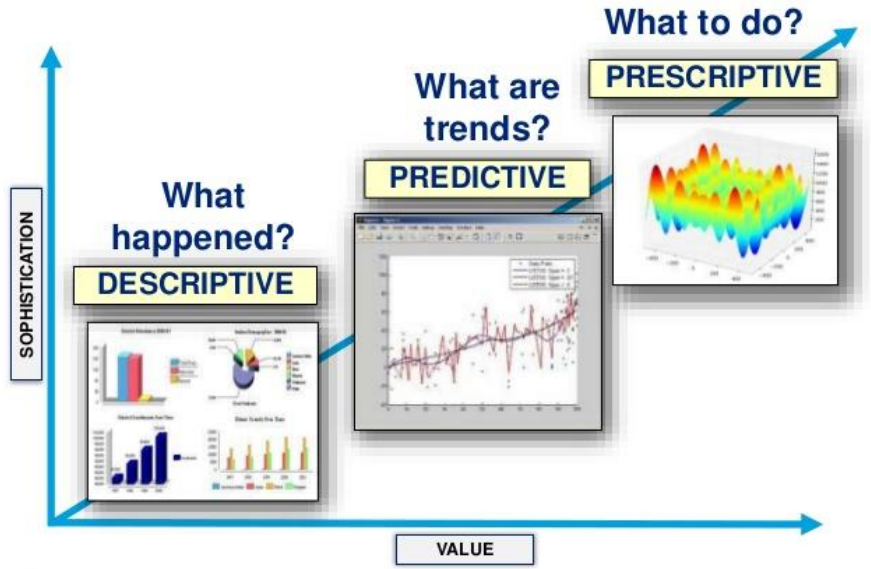
$Y = f(X,T)$
$T^* = \text{argmax}(Y|X)$

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Prescriptive analytics

Differences with descriptive & predictive analytics?

- Incorporates the decision in the analysis
  - Which in descriptive & predictive analytics is left outside the analysis
- Which **directly** allows to optimize decision
  - Decision automation
  - Decision support



Concluding remarks

- Case: Health – personalized medicine, e.g.:
  - Van der Schaar lab @Cambridge: https://www.youtube.com/watch?v=YQ8HX4T5OuE
  - Generative AI and **Virtual Cells** for **Drug Response Modeling**: https://icml.cc/virtual/2025/51534
- Case: Learning analytics, e.g.:
  - https://www.sciencedirect.com/science/article/pii/S0167923620300750
- Case: HR Management, e.g.:
  - Career development, turnover, illness, …
- Case: Fraud risk management?
  - Preventive fraud measures?
- …

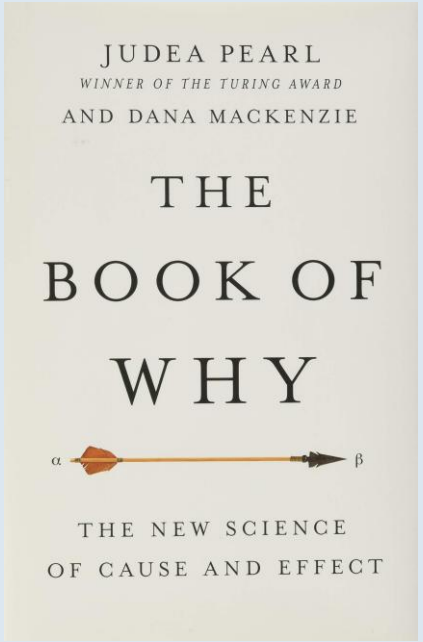
Concluding remarks

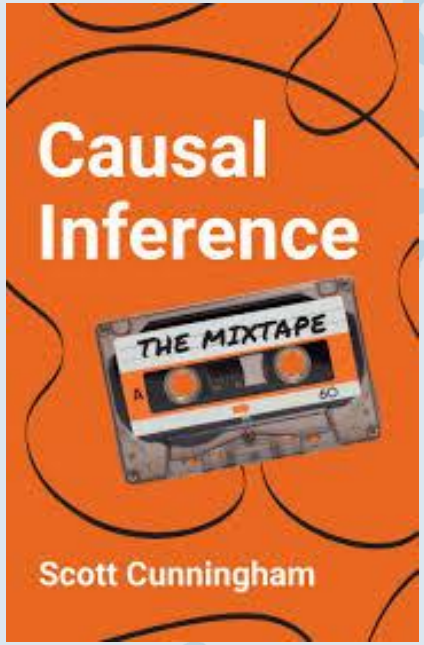
- Advantages:
  - Higher returns: measurable
  - More power (… to do more harm?)
  - Inherently conceptually sound
  - Inherently more interpretable (?)
  - …?
- Disadvantages:
  - Investment costs: data, knowledge, …
  - More complexity
  - …?

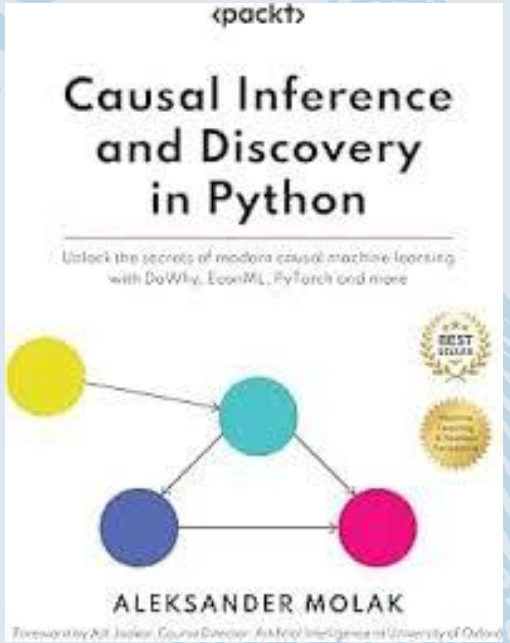
"Everything should be made as simple as possible, but no simpler."
Albert Einstein

"It can scarcely be denied that the supreme goal of all theory is to make the irreducible basic elements as simple and as few as possible without having to surrender the adequate representation of a single datum of experience."

Q&A


JUDEA PEARL
WINNER OF THE TURING AWARD
AND DANA MACKENZIE
THE
BOOK OF
WHY
THE NEW SCIENCE
OF CAUSE AND EFFECT


Causal
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Scott Cunningham


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